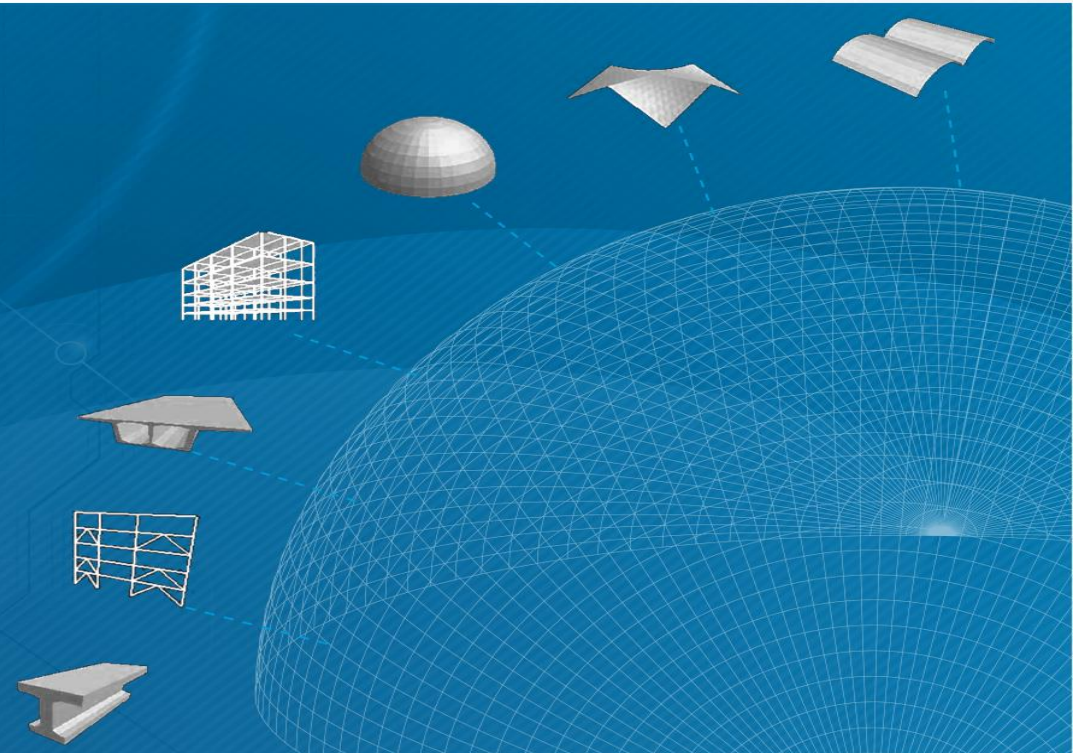
CE – 412 Design of Concrete Structures

Lecture 2:

Materials



October - 2020

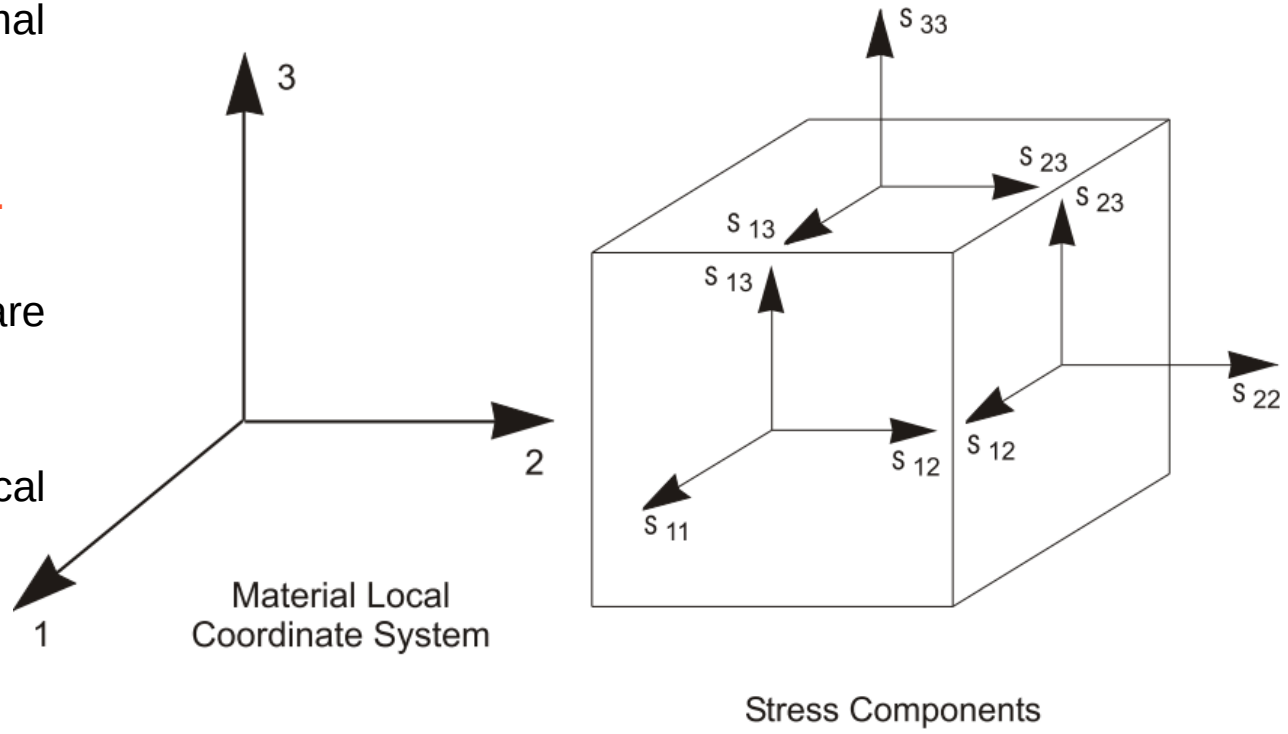
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Basics of Material Behaviour

Stresses and Strains – General Notation

- Each material has its own material local coordinate system, which is used to define the elastic and thermal properties.
- Significant only for orthotropic and anisotropic materials.
- The axes of the material local coordinate system are denoted as 1, 2, and 3.
- The material coordinate system is aligned with the local coordinate system for each element.



Definition of Stress Components in the Material Local Coordinate System

Stress and Strain – One Dimensional Case

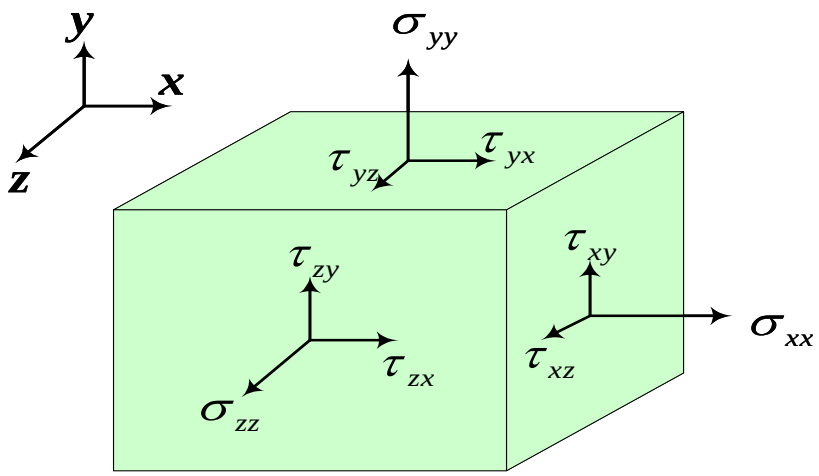
- The Hook's law states that within the elastic limits, the stress is proportional to the strain

$$\sigma = E\,\varepsilon$$

- This is valid for only **very limited cases**
- In reality, the Modulus of Elasticity, **E is NOT a constant**
- There are many stress and strain components, and many properties.

A Bigger Picture of Stress-Strain Components

For a general 3D Isotropic Material



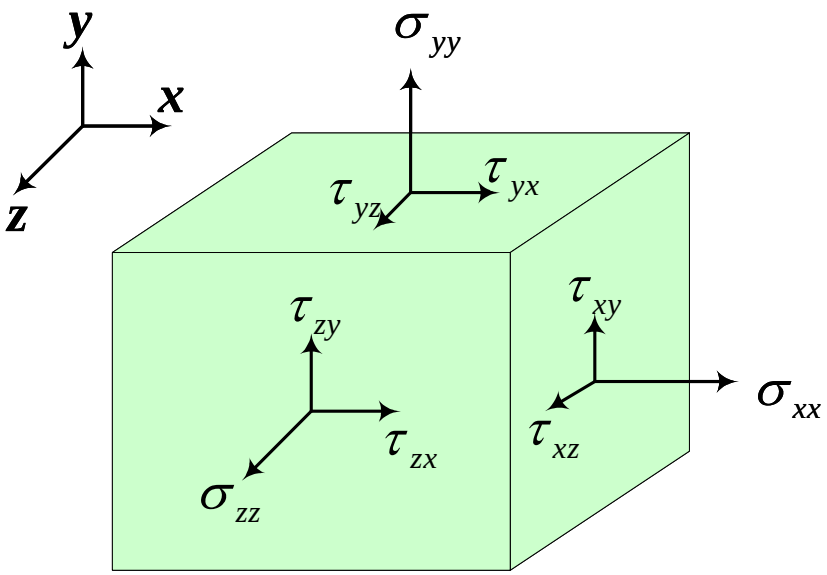
At any point in a continuum, or solid, the stress state can be completely defined in terms of six stress components and six corresponding strains.

Generalized Form of Hook's Law

$$\begin{bmatrix}\sigma_x\\ \sigma_y\\ \sigma_z\\ \tau_{xy}\\ \tau_{yz}\\ \tau_{zx}\end{bmatrix} = \frac{E}{(1+\nu)(1-2\nu)}\begin{bmatrix}1-\nu & \nu & \nu & 0 & 0 & 0\\ \nu & 1-\nu & \nu & 0 & 0 & 0\\ \nu & \nu & 1-\nu & 0 & 0 & 0\\ 0 & 0 & 0 & \frac{1-2\nu}{2} & 0 & 0\\ 0 & 0 & 0 & 0 & \frac{1-2\nu}{2} & 0\\ 0 & 0 & 0 & 0 & 0 & \frac{1-2\nu}{2}\end{bmatrix}\begin{bmatrix}\varepsilon_x\\ \varepsilon_y\\ \varepsilon_z\\ \gamma_{xy}\\ \gamma_{yz}\\ \gamma_{zx}\end{bmatrix}$$

Figure 2.12 The relationship between stress and strain in a three-dimensional isotropic material (generalized form of Hook's law with a constant E and ν in all directions).

Stress and Strains Components



At any point in a continuum, or solid, the stress state can be completely defined in terms of six stress components and six corresponding strains.

- The Hook's law is a simplified form of Stress-Strain relationship.
- Ultimately, the six stress and strain components can be represented by 3 principal summations.

Reinforced Concrete

The concrete which contains steel bars to enhance its tensile strength







Advantages and Disadvantages of RC

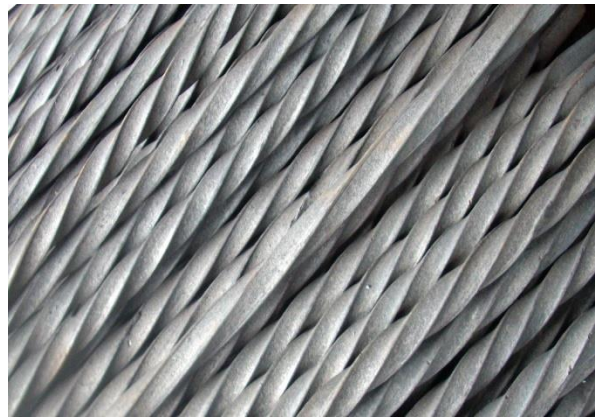
- Considerable compressive strength
- Greater resistance to fire and weathering effects
- Higher stiffness
- Low maintenance material
- Increase in strength with passage of time
- Cost effective
- A lower grade of skilled labor
- Can be casted into different variety of shapes

- Low tensile strength
- Forms (shuttering) are required to hold in place
- Low weight per unit volume causes heavier structural members
- Variation in properties through a member.
- Is very sensitive to hydration – related to strength

The Compatibility of Concrete and Steel

- Both materials are managing the pros and cons of each other like:
  - Low tensile strength of concrete – higher tensile strength of steel
  - Weaker resistance to corrosion for steel – higher weathering resistance of concrete
  - Poor fire resistance to steel – higher insulation properties of concrete
  - Lighter in weight (steel) – heavier unit weight (concrete)
- Ribs in reinforcing bars lead to better bond properties
- Both materials have almost same co-efficient of thermal expansion
  - 0.0000065 per unit length per degree Fahrenheit for steel
  - 0.0000040 to 0.0000070 with average of 0.0000055 for concrete



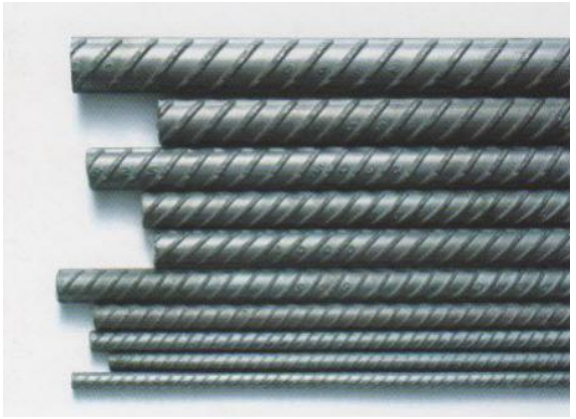


Compressive, Tensile & Shear Strength of RC

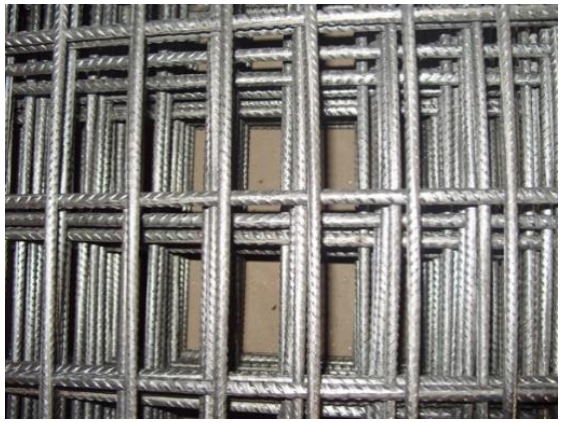
- Compressive strength
  - Determined by 28 days test specimen
  - The strength may vary from 3000 to 6000 psi as normal concrete and beyond 6000 psi as high strength concrete
- Tensile strength
  - Related to opening of micro cracks in concrete
  - Is usually determined from modulus of ruptures or split cylinder test
  - Usually around $1/10^{th}$ of the compressive strength
- Shear Strength
  - Controlled by diagonal tension effect
  - Usually vary from one-third to four-fifth of ultimate compressive strength

Steel Reinforcement

- Reinforcing steel may be hot-rolled or cold-twisted
- May be plain or deformed
- Sizes start from #3 to #11 and for special cases we may refer to #14 and #18 also
- Welded wire mesh can also be used
- Grades of steel may vary from 40 to 70 with 60 as most common







Basic Properties of Concrete and Steel

- For analysis
  - Modulus of elasticity, E
  - Poisons ratio, ν
  - Shear modulus, G
  - Thermal expansion coefficient, α

Initial Modulus

Tangent Modulus

Secant Modulus

Long term Modulus

- For design
  - Yield stress, F_y
  - Failure stress, F_u, F_c, F_t etc.
  - Yield strain ε_y
  - Failure strain ε_u

Material Modelling for Steel and Concrete

- Modulus of elasticity based on nominal strengths
- Unit weight of materials
- Unit mass of materials
- Poisson's ratios
- Coefficient of Thermal Expansion (Concrete = 0.0000055 1/F, Steel = 0.0000065 1/F)

Modulus of elasticity of concrete,

$$E_c = 57000\sqrt{f_c'} \text{ for } f_c' \leq 6000 \text{ psi}$$

$$E_c = 4700\sqrt{f_c'} \text{ for } f_c' \leq 40 \text{ MPa}$$

$$E_c = 40000\sqrt{f_c'} + 1 \times 10^6 \text{ for } f_c' > 6000 \text{ psi}$$

Modulus of elasticity of steel,

$$E_s = 199{,}947 \text{ MPa} = 29{,}000 \text{ ksi}$$

Weight per unit volume

- Reinforced concrete = 23.56 kN/m^3 (150 lb/ft^3)
- Steel = 76.97 kN/m^3 (490 lb/ft^3)

Poisson's ratio

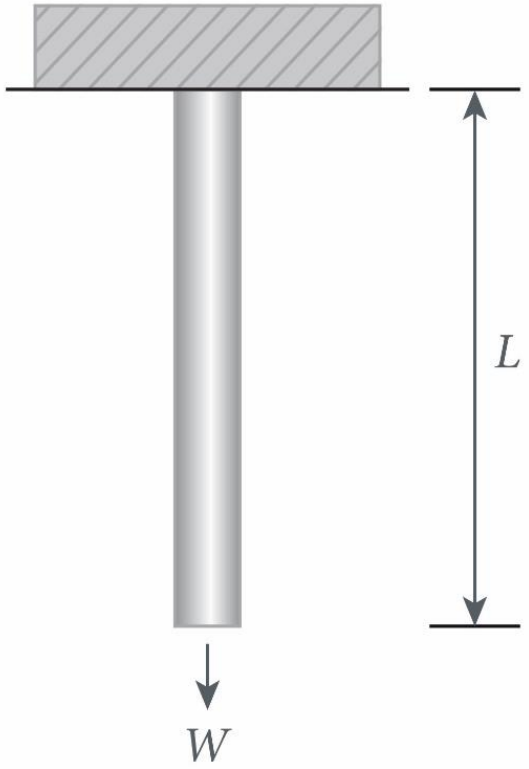
- Reinforced concrete = 0.2
- Steel = 0.3

Other Specific properties

- Relaxation
- Fatigue
- Creep
- Shrinkage
- Confinement based properties

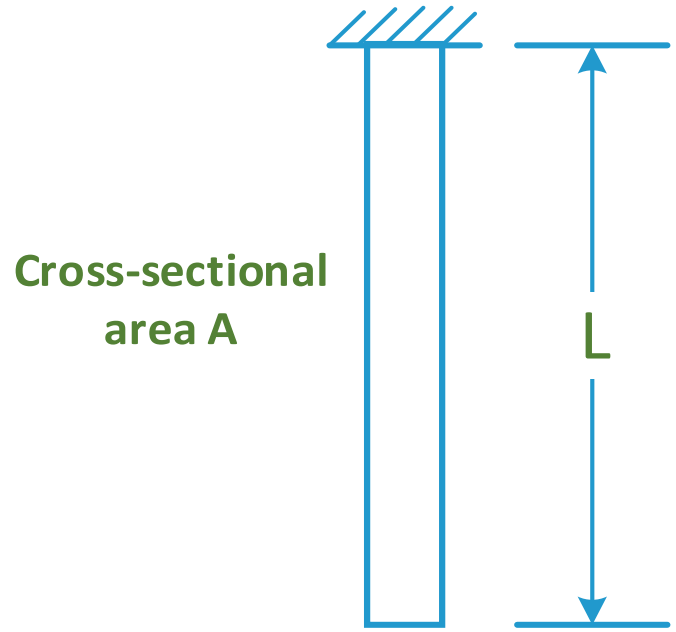
The Concept of Specific Length

$$\text{Total mass} = m$$
$$\text{Cross-section Area} = A$$
$$\text{Density of Material} = \rho$$
$$\text{Failure Stress of Material} = \sigma_F$$
$$\text{if}$$
$$\text{Total Weight} = \text{Failure Load}$$
$$m\,g = \sigma_F\,A$$
$$\rho\,L\,A\,g = \sigma_F\,A$$
$$\text{Hence, Specific Length}$$
$$L = \sigma_F\,/\rho\,g$$



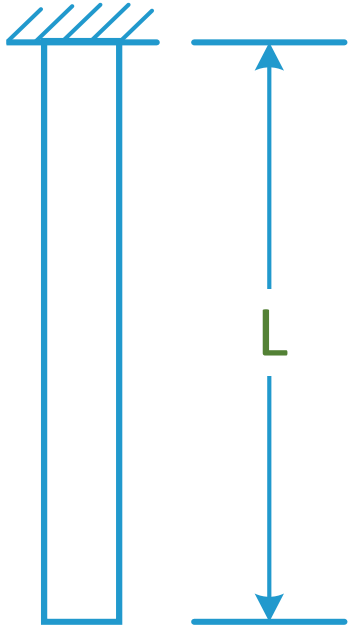
The Concept of Specific Strength

- First to realize this: Galileo Galilei (1654AD – 1642 AD)
- All structures have to support their own weight
- Can the size of a structure be increased indefinitely for it to be able to carry its own weight?
- Problem: how long a bar of uniform cross-section can be before it breaks due to its own weight?
- Equate the weight of the bar to its tensile strength:
    - Weight = Tensile resistance



The Concept of Specific Strength

- Weight = Volume × specific weight
  - $W = A \times L \times \rho \times g$
- Tensile resistance = Area × Ultimate tensile strength
  - $R = A \times Tu$
- Equate weight to resistance:
  - $W = R$ $A \times L \times \rho \times g = A \times Tu$
  - $L = Tu / (r \times g) = S$ = specific strength



- There is an absolute limit (= S) to the length that the bar can attain without breaking
- Larger a structure is, larger is the proportion of its own weight to the total load that can be carried by itself

The Concept of Specific Strength

- For structures subjected to tension/compression, as the size of an object increases, its strength increases with the square of the ruling dimensions, while the weight increases with its cube.
- For each type of structure there is a maximum possible size beyond which it cannot carry even its own weight
- Consequences:
  - it is impossible to construct structures of enormous size
  - there is a limit to natural structures (trees, animals, etc.)
  - larger a structure becomes, stockier and more bulky it gets
    - large bridges are heavier in proportions than smaller ones
    - bones of elephants are stockier and thicker than the ones of mice
  - proportions of aquatic animals are almost unaffected by their size (weight is almost entirely supported by buoyancy).

Ultimate and Specific Strengths

Material			Ultimate strength σ_u (N/mm^2)		Specific strength S (m)	
			Tensile	Compression	Tensile	Compression
Traditional	Stone	Granite	40	200	1400	7000
		Limestone	5	40	225	1800
	Brick		6	60	320	3200
	Timber (spruce)	Along grain	120	30	24000	6000
		Across grain	3.5	-	700	-
Modern	Portland cement concrete	Normal use	2	20	90	900
		High strength	6	60	270	2700
	Iron and steel	Mild steel	355	355	4500	4500
		High strength steel	600	600	8000	8000
		Very high-strength prestressing wires	2000	-	26700	-
	Aluminium alloy (dural)		450	450	17000	17000

Specific Strength

- Stone, brick and concrete: used in compression
- Steel: used in tension
- Timber: excellent performance in terms of specific strength, especially in tension
- Aluminum: high specific strength
- Aircrafts must carry loads and must be capable of being raised into the air under their own power materials with high specific strength
    - wood was extensively used in early planes
    - modern material: aluminum

Structural materials: Stiffness and Ductility

Material			Modulus of elasticity E (N/mm^2)	Specific modulus M (m $\times 10^5$)	Ductility
Traditional	Stone	Granite	45 000	1.57	Brittle
		Limestone	30 000	1.35	
	Brick		30 000	1.60	
	Timber (spruce)	Along grain	15 000	3.00	NA
		Across grain	-	-	
Modern	Portland cement concrete	Normal use	25 000	1.12	Brittle
		High strength	40 000	1.80	
	Iron and steel	Mild steel	210 000	2.80	Large ductility
		High strength steel	210 000	2.80	Moderate ductility
		Very high-strength prestressing wires	210 000	2.80	Low ductility
	Aluminium alloy (dural)		70 000	2.80	Ductile

Directional Behaviors of Materials

Isotropic Materials

- The behavior of an isotropic material is independent of the direction of loading or the orientation of the material.
- Shearing behavior is uncoupled from ex-tensional behavior and is not affected by temperature change.

Orthotropic Materials

- The behavior of an orthotropic material can be different in each of the three local coordinate directions.
- Shearing behavior is un-coupled from extensional behavior and is not affected by temperature change.

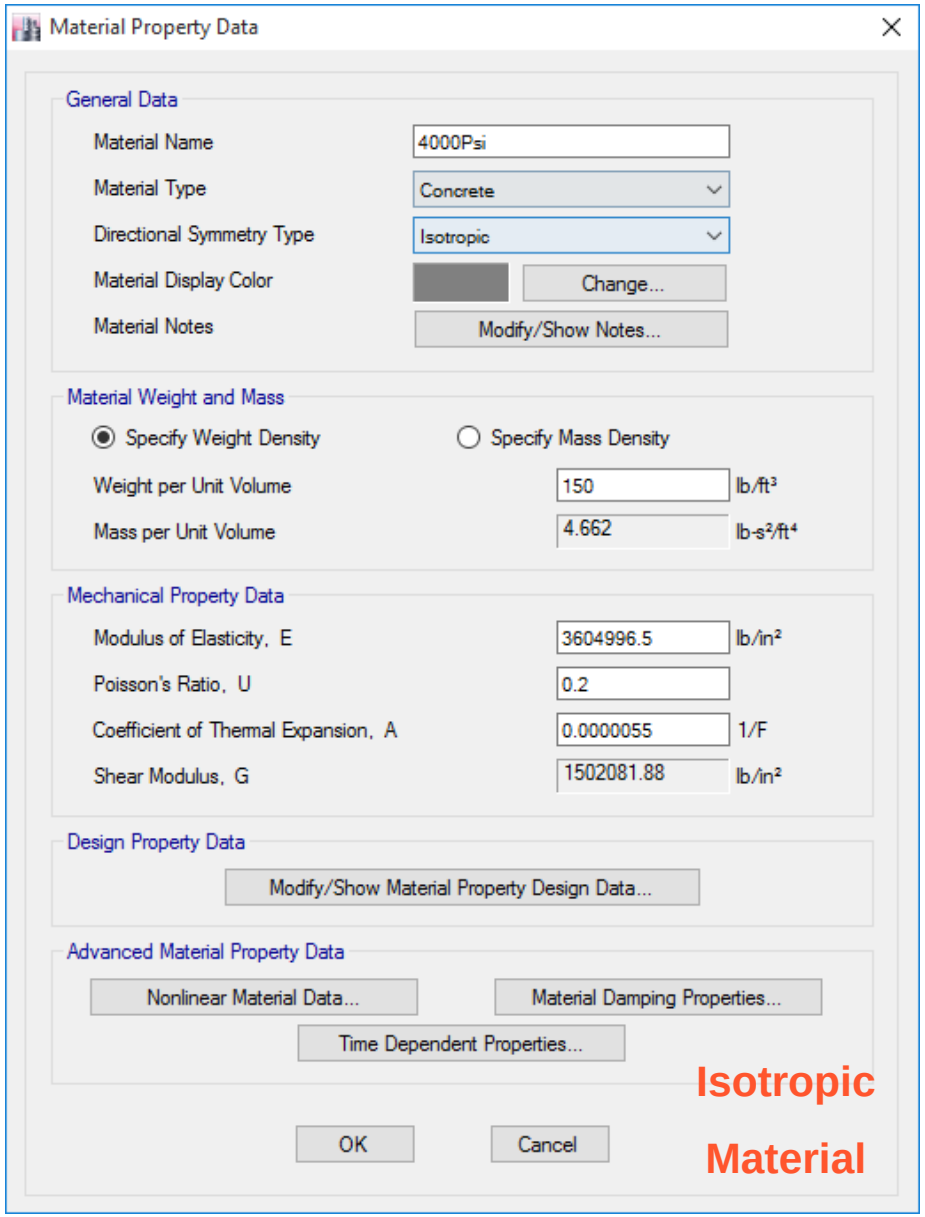
Anisotropic Materials

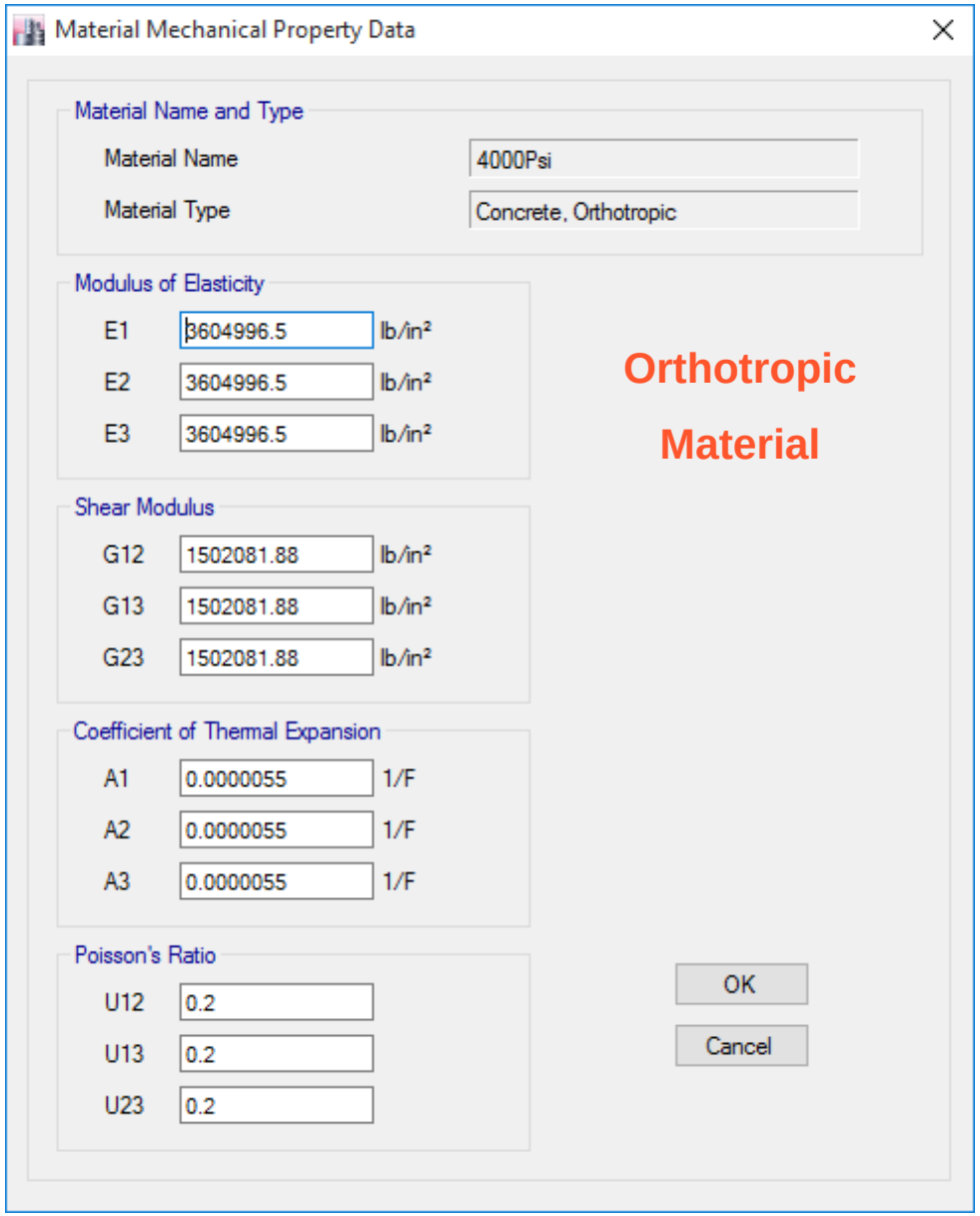
- The behavior of an anisotropic material can be different in each of the three local coordinate directions
- Shearing behavior can be fully coupled with extensional behavior and can be affected by temperature change.

Directional Behavior of Materials

Table 2.1 Directional Behavior-Based Classification of Materials

Isotropic	• Same behavior in all directions • All directions uncoupled	• Steel • Material used in line type of members (beams, columns, etc.)
Orthotropic	• Different behavior in orthogonal directions • Behavior is uncoupled	Metal deck, reinforced concrete slabs
Anisotropic	• Different behavior in three directions • Behavior is coupled	Soil, some types of rubber


Material Property Data
General Data
Material Name
4000Psi
Material Type
Concrete
Directional Symmetry Type
Isotropic
Material Display Color
Change...
Material Notes
Modify/Show Notes...
Material Weight and Mass
Specify Weight Density
Specify Mass Density
Weight per Unit Volume
150
lb/ft³
Mass per Unit Volume
4.662
lb-s²/ft⁴
Mechanical Property Data
Modulus of Elasticity, E
3604996.5
lb/in²
Poisson's Ratio, U
0.2
Coefficient of Thermal Expansion, A
0.0000055
1/F
Shear Modulus, G
1502081.88
lb/in²
Design Property Data
Modify/Show Material Property Design Data...
Advanced Material Property Data
Nonlinear Material Data...
Material Damping Properties...
Time Dependent Properties...
OK
Cancel
Isotropic
Material

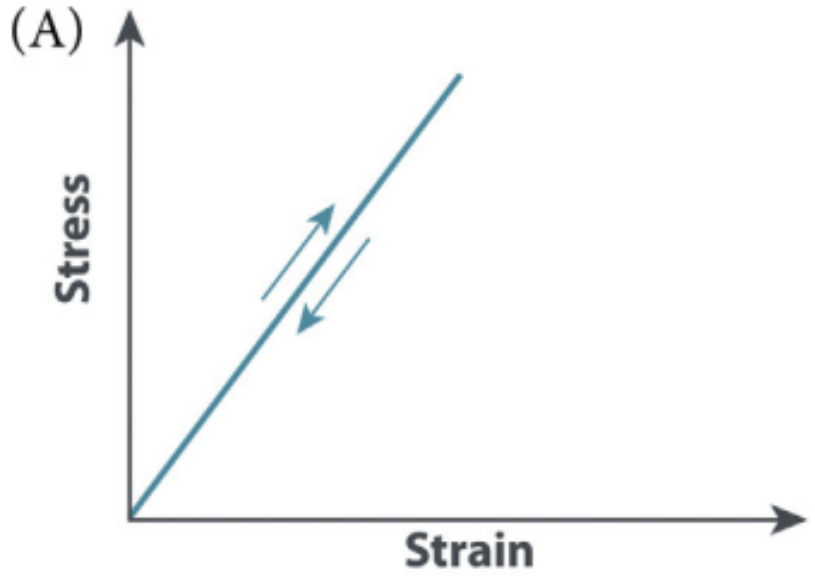

Material Mechanical Property Data
Material Name and Type
Material Name
4000Psi
Material Type
Concrete, Orthotropic
Modulus of Elasticity
E1
3604996.5
lb/in²
E2
3604996.5
lb/in²
E3
3604996.5
lb/in²
Shear Modulus
G12
1502081.88
lb/in²
G13
1502081.88
lb/in²
G23
1502081.88
lb/in²
Coefficient of Thermal Expansion
A1
0.0000055
1/F
A2
0.0000055
1/F
A3
0.0000055
1/F
Poisson's Ratio
U12
0.2
U13
0.2
U23
0.2
OK
Cancel
Orthotropic
Material

Linearity and Elasticity

- Material behavior depends on level of strain
  - Linear
  - Non-linear
- Material behavior depends on loading history
  - Elastic
  - Plastic
  - Inelastic
  - Hysteretic

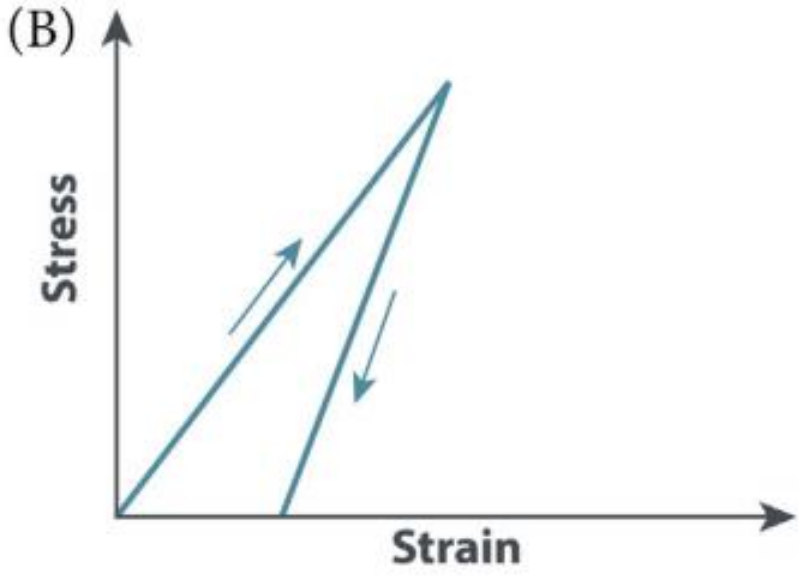
Linear Elastic Material

- A linear elastic material is one in which the strain is proportional to stress
- Both “loading” and “unloading” curves are same (straight lines).



Linear Inelastic Material

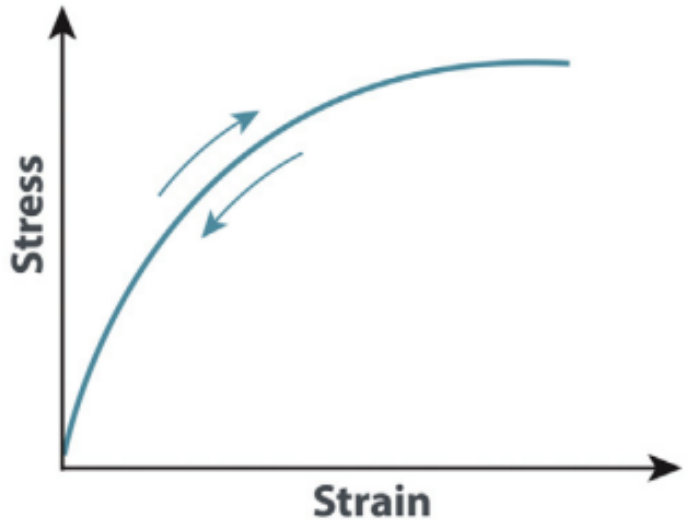
- A linear inelastic material is one in which the strain is proportional to stress
- “Loading” and “unloading” curves are not same (although straight lines).



Nonlinear Elastic material

- For a nonlinear elastic material, strain is not proportional to stress as shown in figure.
- Both “loading” and “unloading” curves are same but are not straight lines.

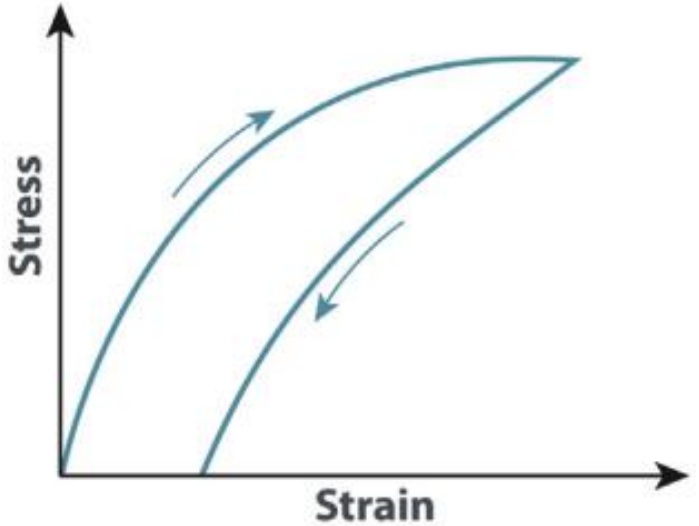
(C)



Nonlinear Inelastic Material

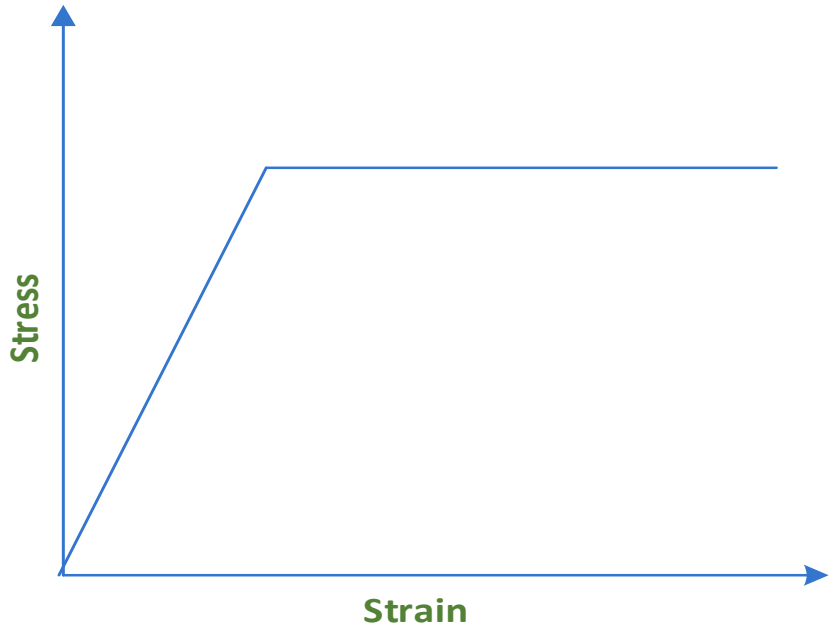
- For a nonlinear inelastic material, strain is not proportional to stress as shown in figure.
- “Loading” and “unloading” curves are not same in this case.

(D)



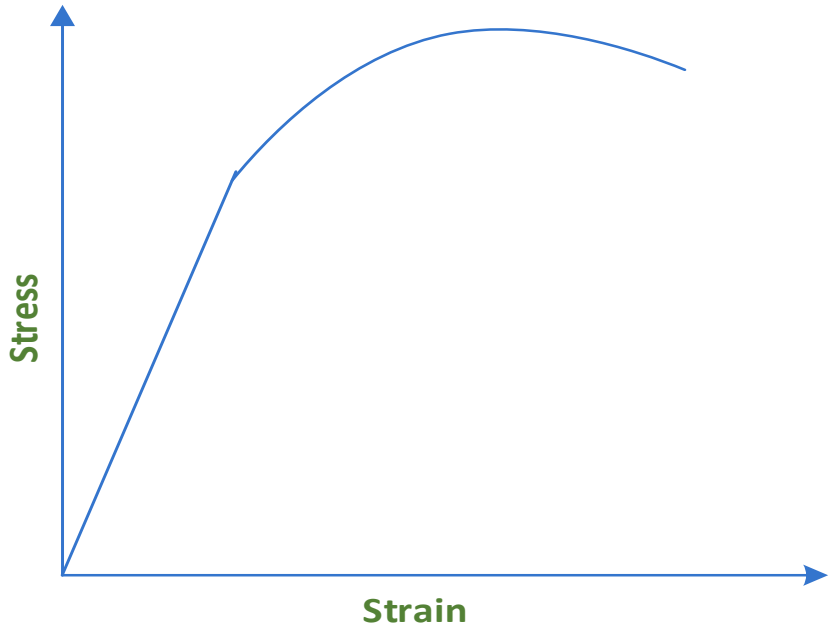
Elastic–Perfectly Plastic (Non-strain Hardening)

- The behavior of an elastic-perfectly plastic (non-strain hardening) material



Elastic – Plastic Material

- The elastic plastic material exhibits a stress – strain behavior as depicted in the figure



Dependence of Behavior

- Relationship between **Stress and Strain** depends on
  - Basic material composition
  - Initial conditions
  - State of strain
  - Direction of strain
  - History of strain
  - Time since initial strain
  - Temperature
  - Cyclic strain
  - Rate of strain change, velocity and acceleration

Stress-strain Curves of Concrete and the Concept of Ductility

Typical Stress-strain Curves

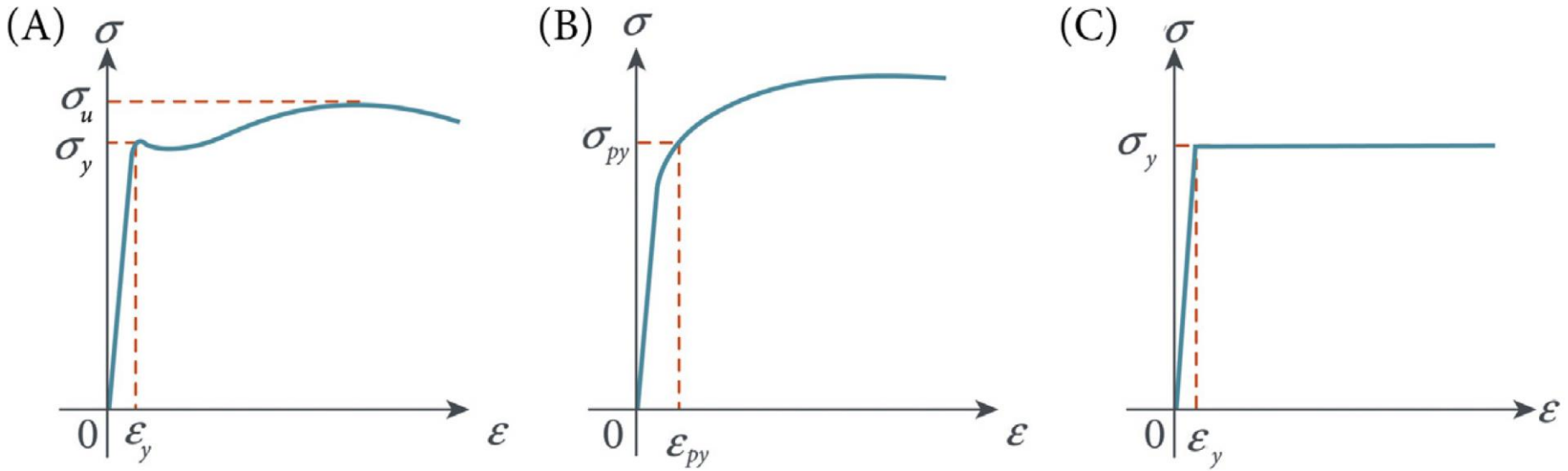


Figure 2.16 Typical stress–strain curves for (A) mild steel, (B) prestressing steel, and (C) simplified curve.

Typical Stress-strain Curves

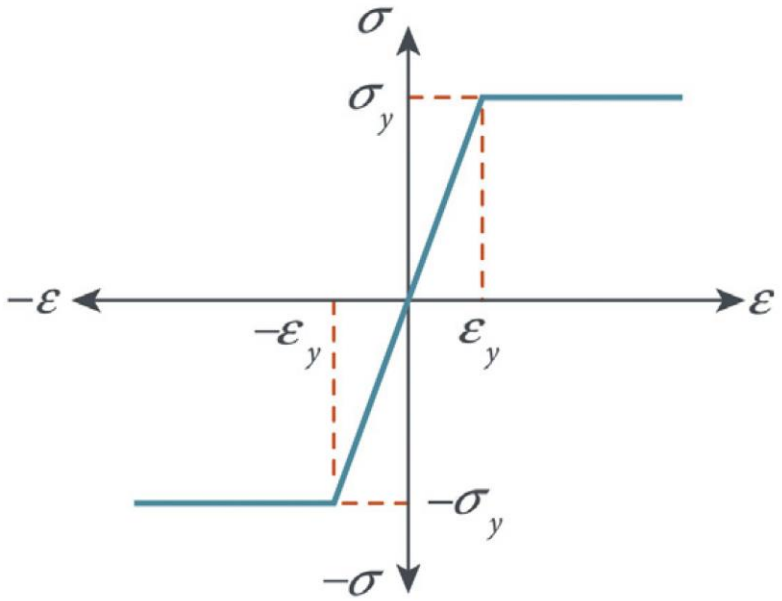


Figure 2.17 An idealized complete elasto-plastic stress−strain curve for steel.

Typical Stress-strain Curves

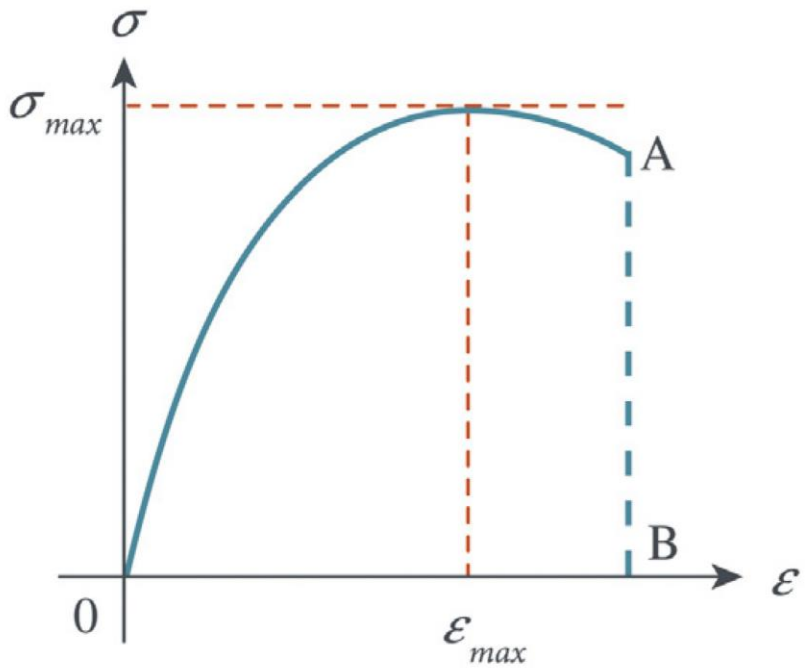
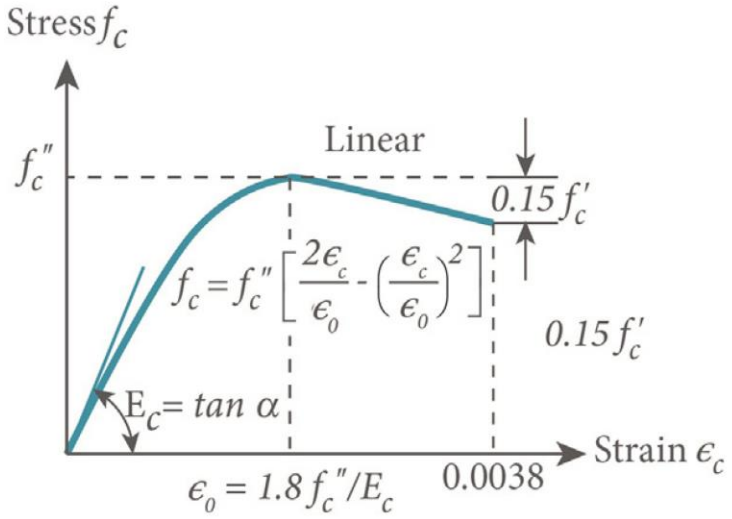
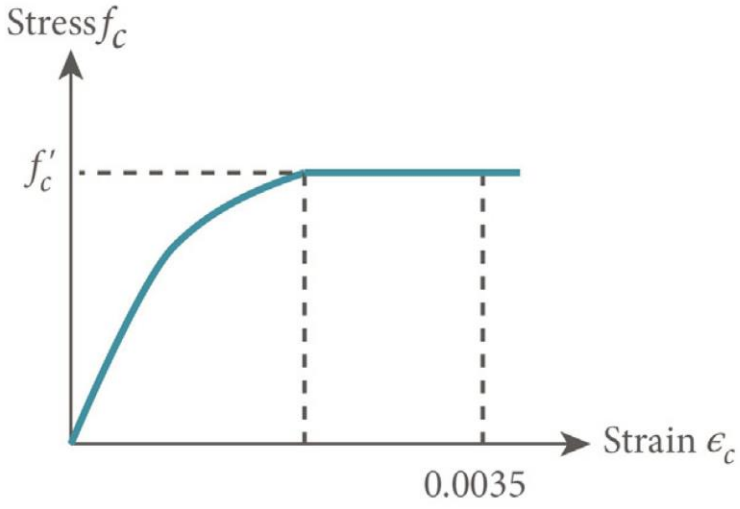


Figure 2.18 Typical stress–strain curve for concrete.

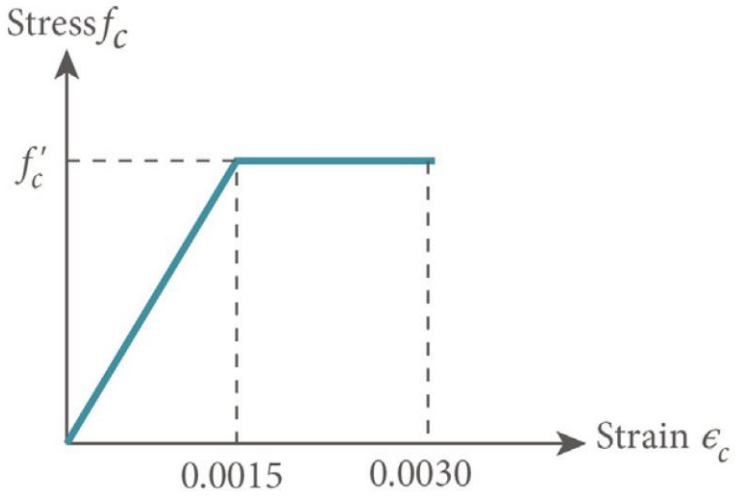
Some Examples of Stress-strain Models for Reinforced Concrete



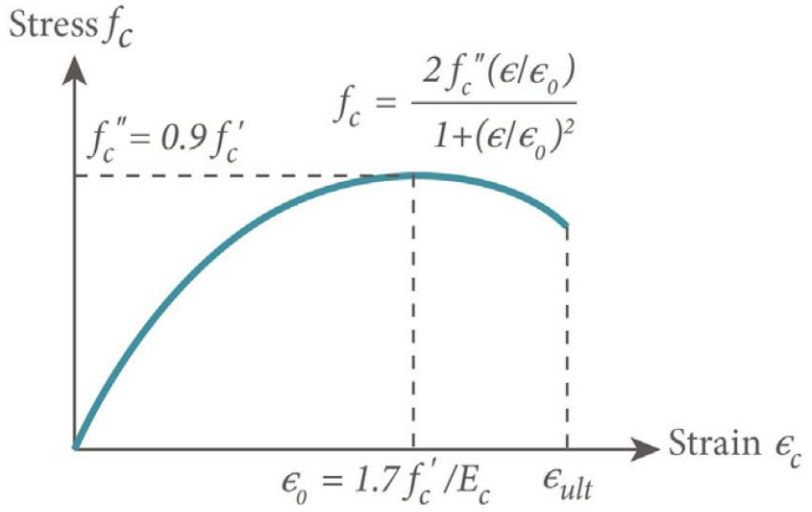
(A) Modified hognestad



(B) Parabola rectangle



(C) Elastic-plastic



(D) Todeschini model

Various stress-strain Models for Unconfined Concrete

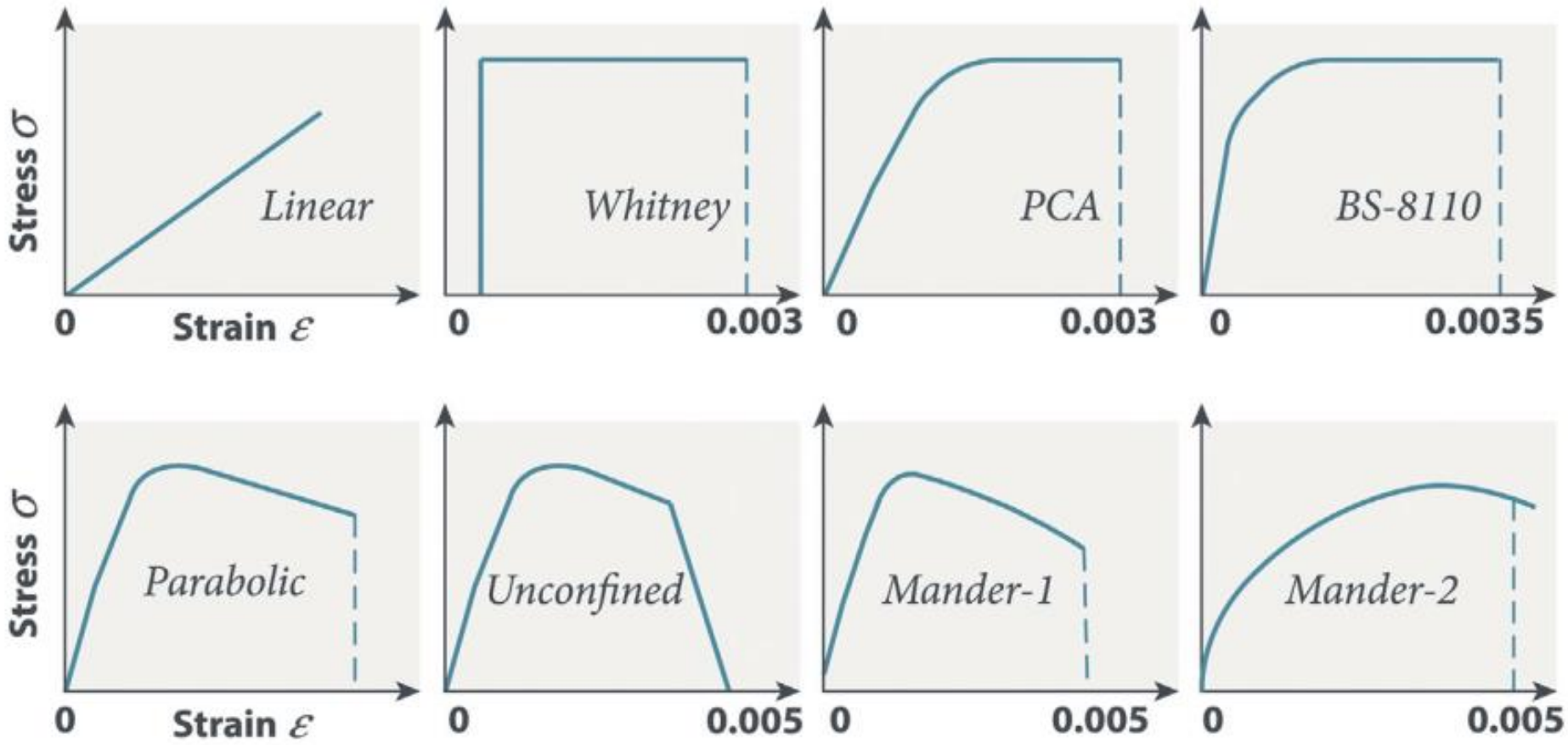
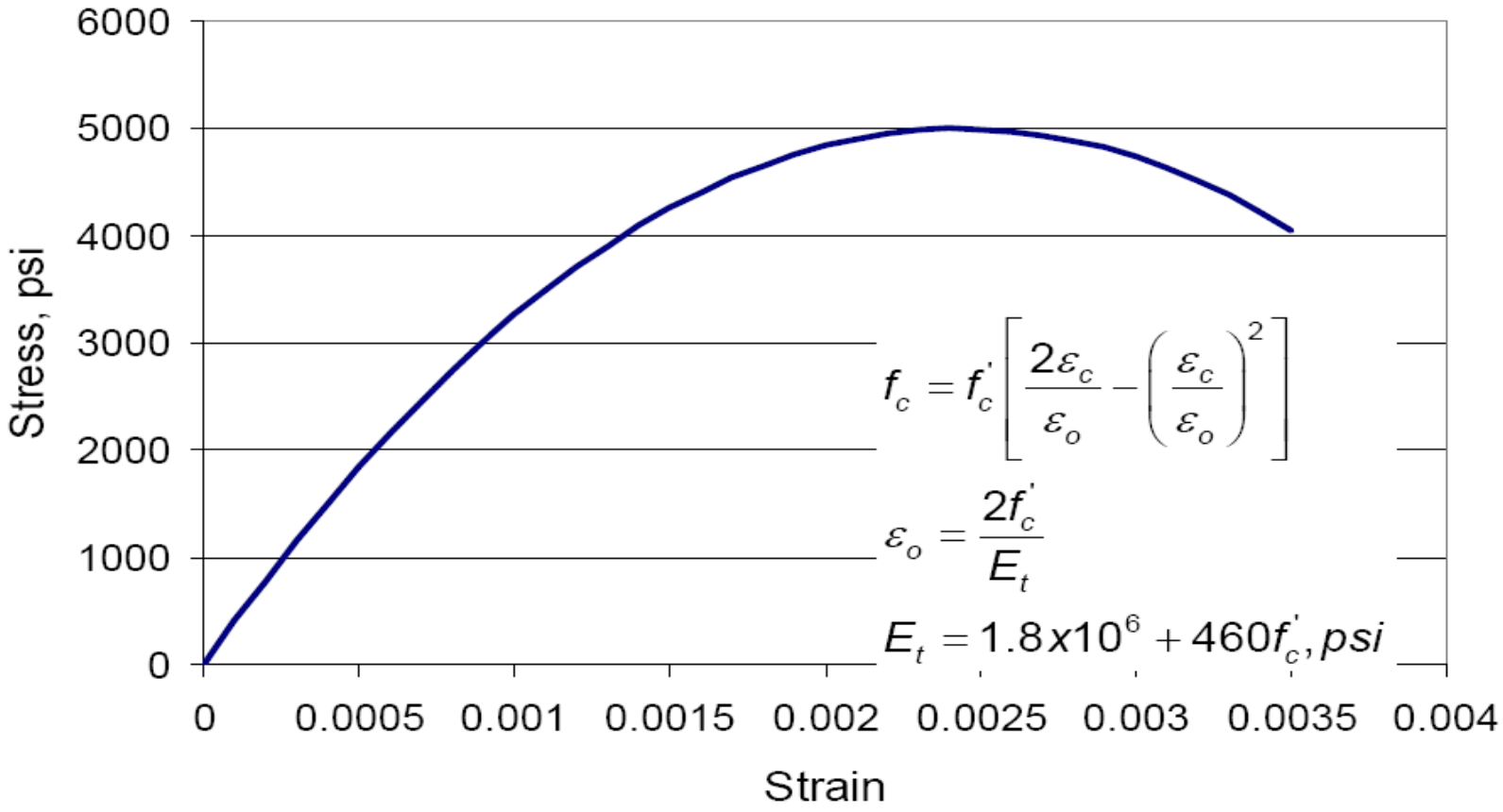


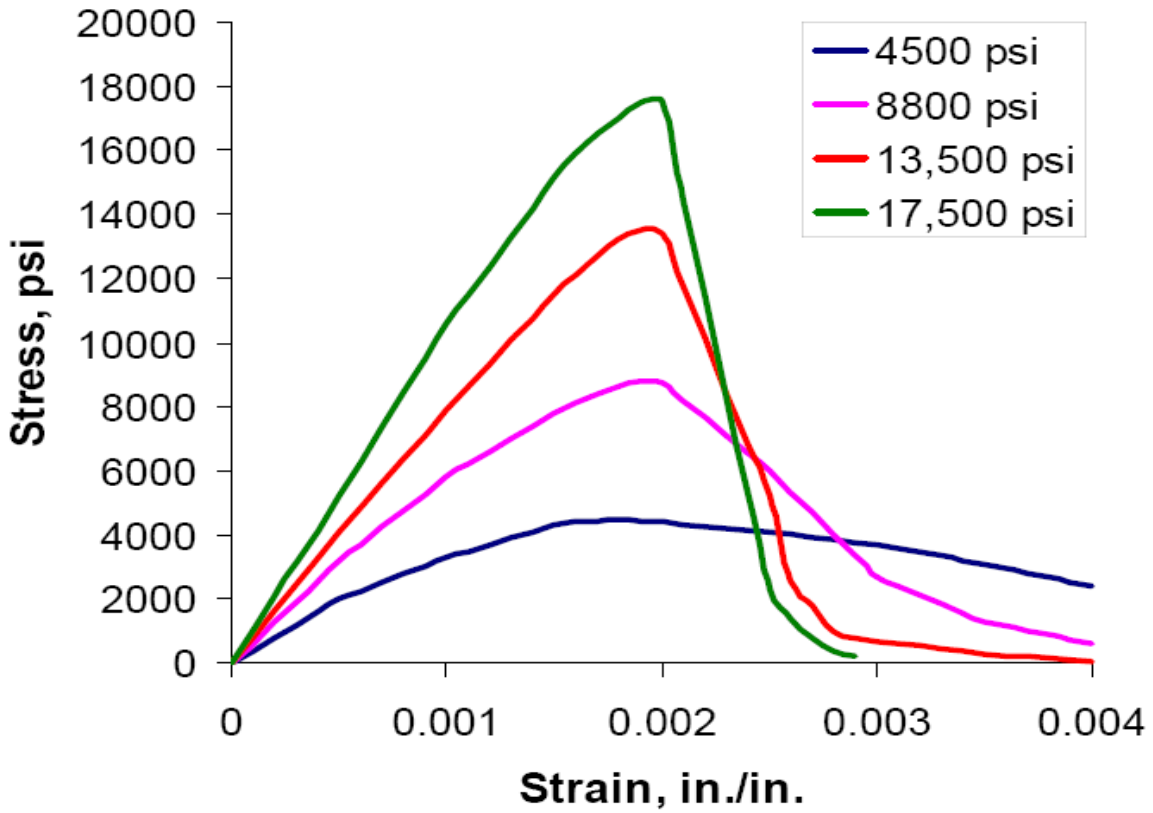
Figure 2.20 Symbolic comparison of various stress—strain curves for concrete.

An Example: Hognestad's Parabola



Idealized Stress-Strain Behavior of Unconfined Concrete

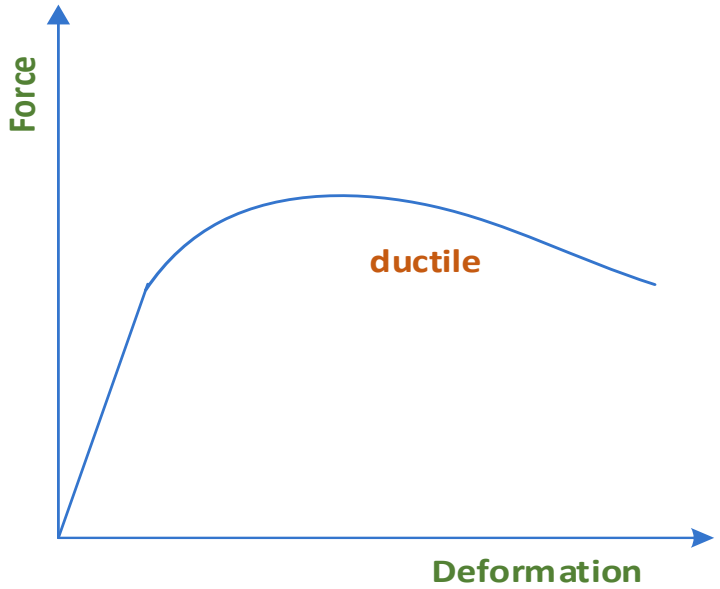
Concrete Behavior

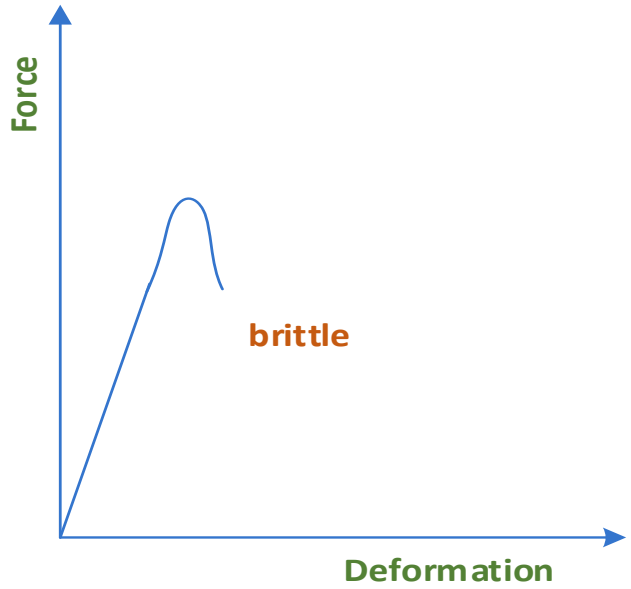


Unconfined Concrete Stress-Strain Behavior

Ductile and Brittle Materials

- Ductile materials:
  - able to deform significantly into the inelastic range
- Brittle materials:
  - fail suddenly by cracking or splintering
  - much weaker in tension than in compression





Structural materials: Ductility

- Ductility is important for the "ultimate" behavior of structures.
- Most structures are designed to respond in the elastic range under service loads, but, given the uncertainties in real strength of material, behavior of the structure, magnitude of loading, and accidental actions, a structure can be subjected to inelastic deformations.
- A ductile material will sustain large deformations before collapsing, "warning" the people inside.
- A ductile material allows for redistribution of stresses in statically indeterminate structures, which are able to support larger loads than in the case of a structure realized of brittle material.

Ductile and Brittle Materials

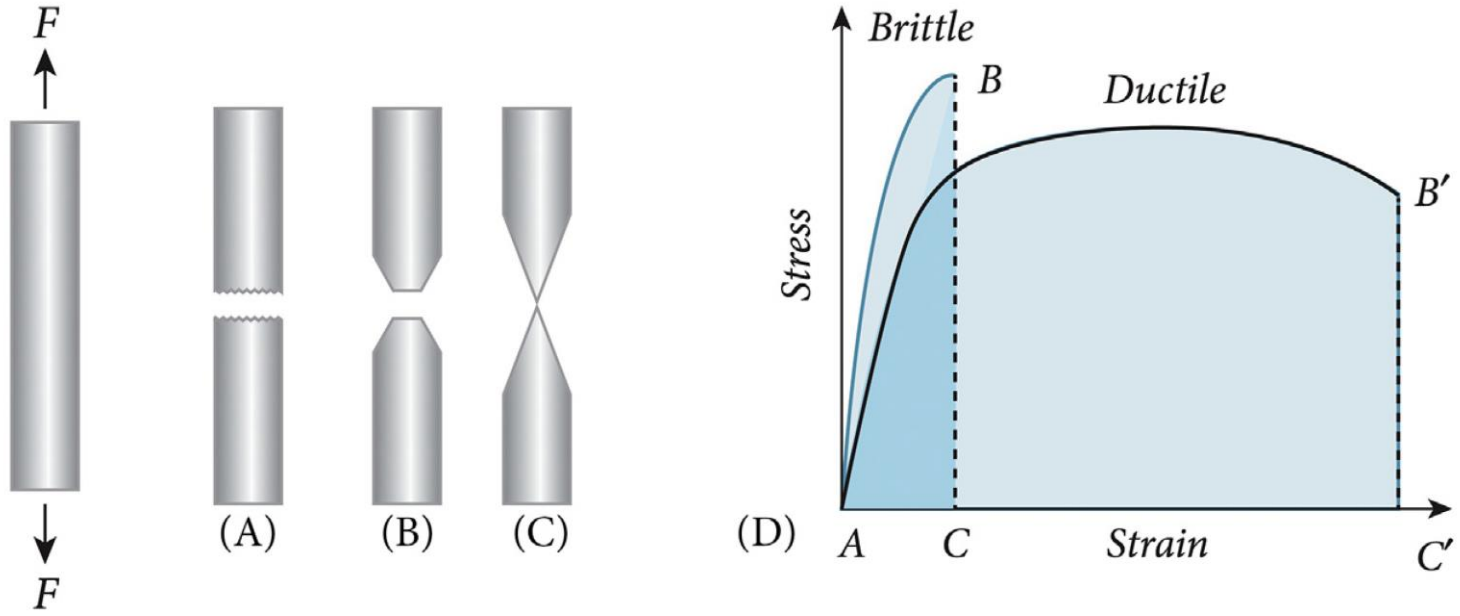


Figure 2.14 Some brittle and ductile material behavior. (A) brittle fracture; (B) ductile fracture; (C) completely ductile failure; (D) typical stress–strain curves of brittle (ABC) and ductile (A′B′C′) materials.

Ductility Ratio

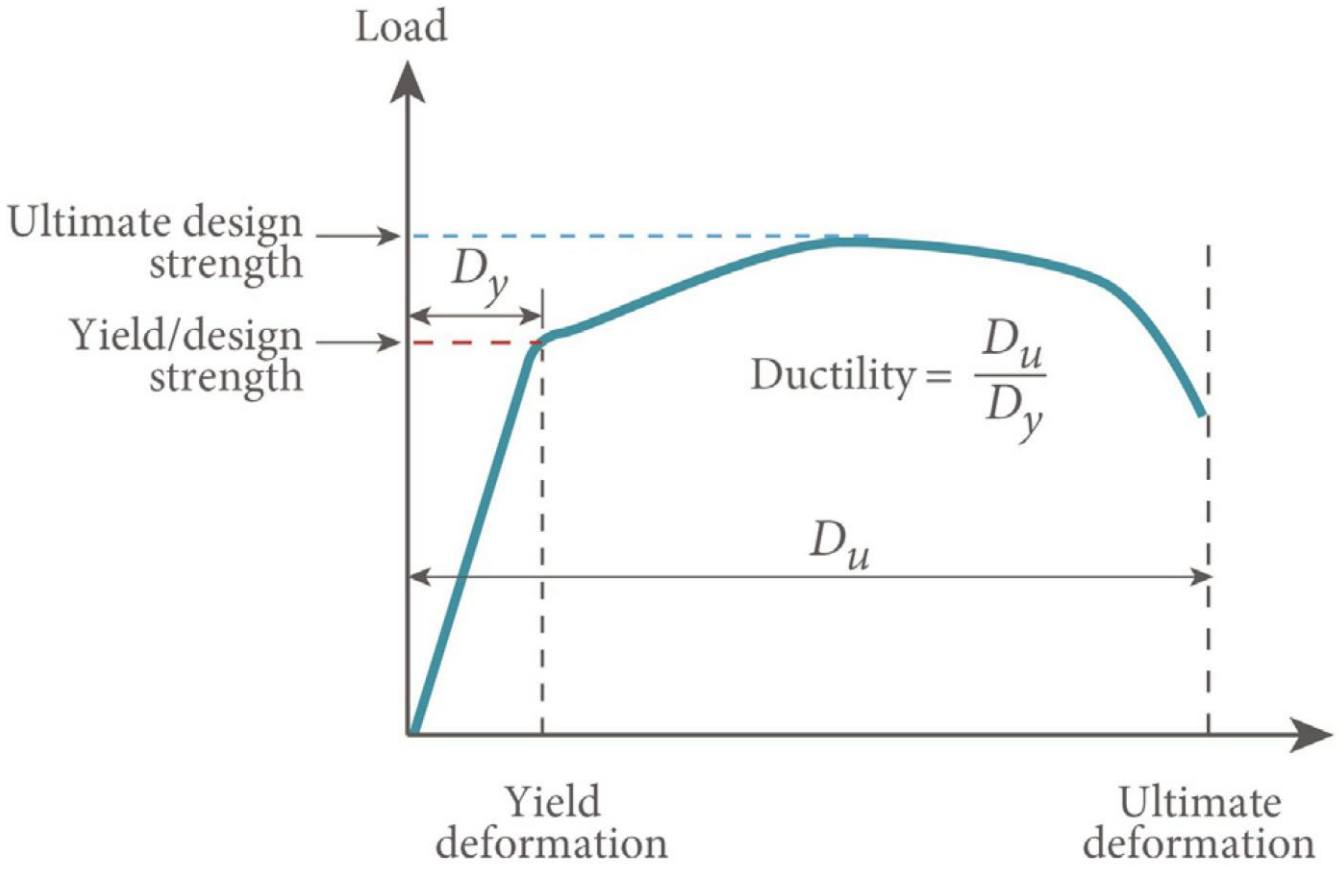


Figure 6.1 The definition of ductility ratio, $\mu = D_u/D_y$.

What is Global (Structural) Ductility “made” of ???

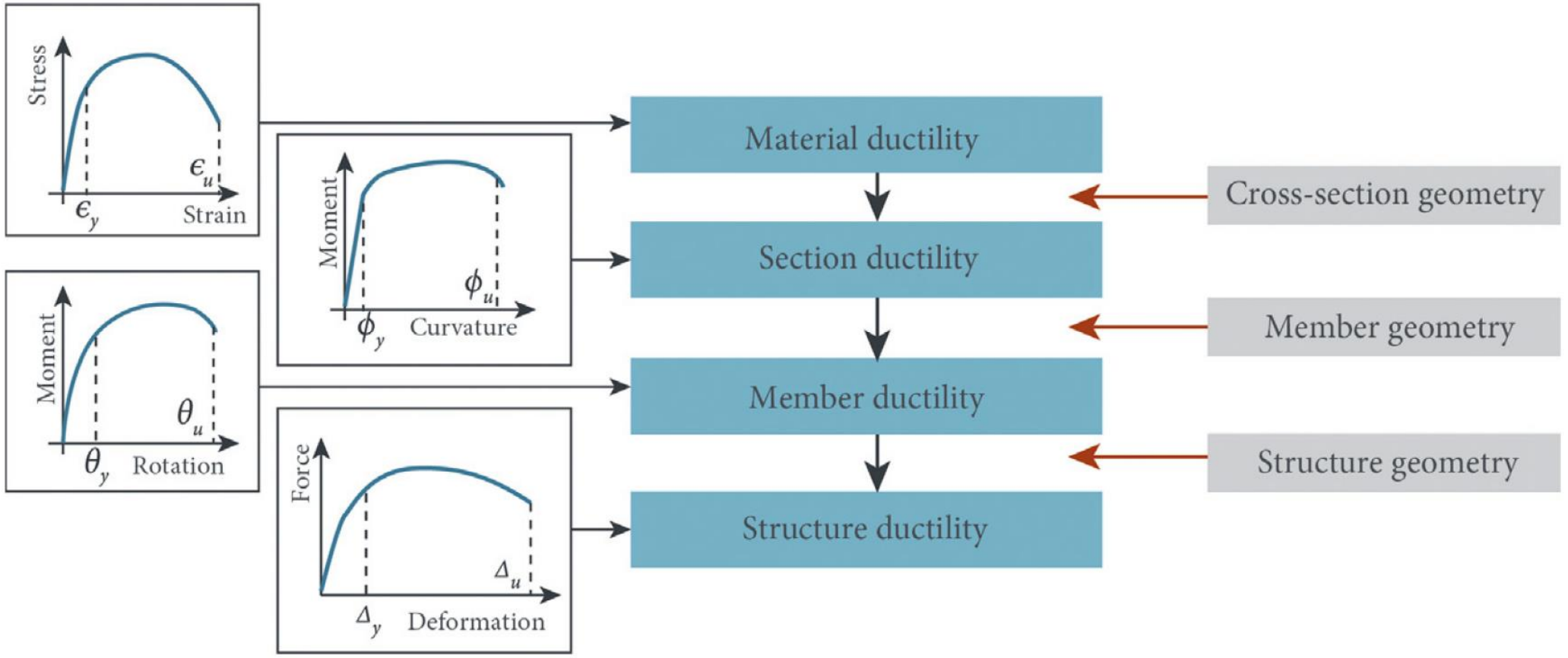


Figure 6.2 Various levels of ductility.

Confinement and Ductility

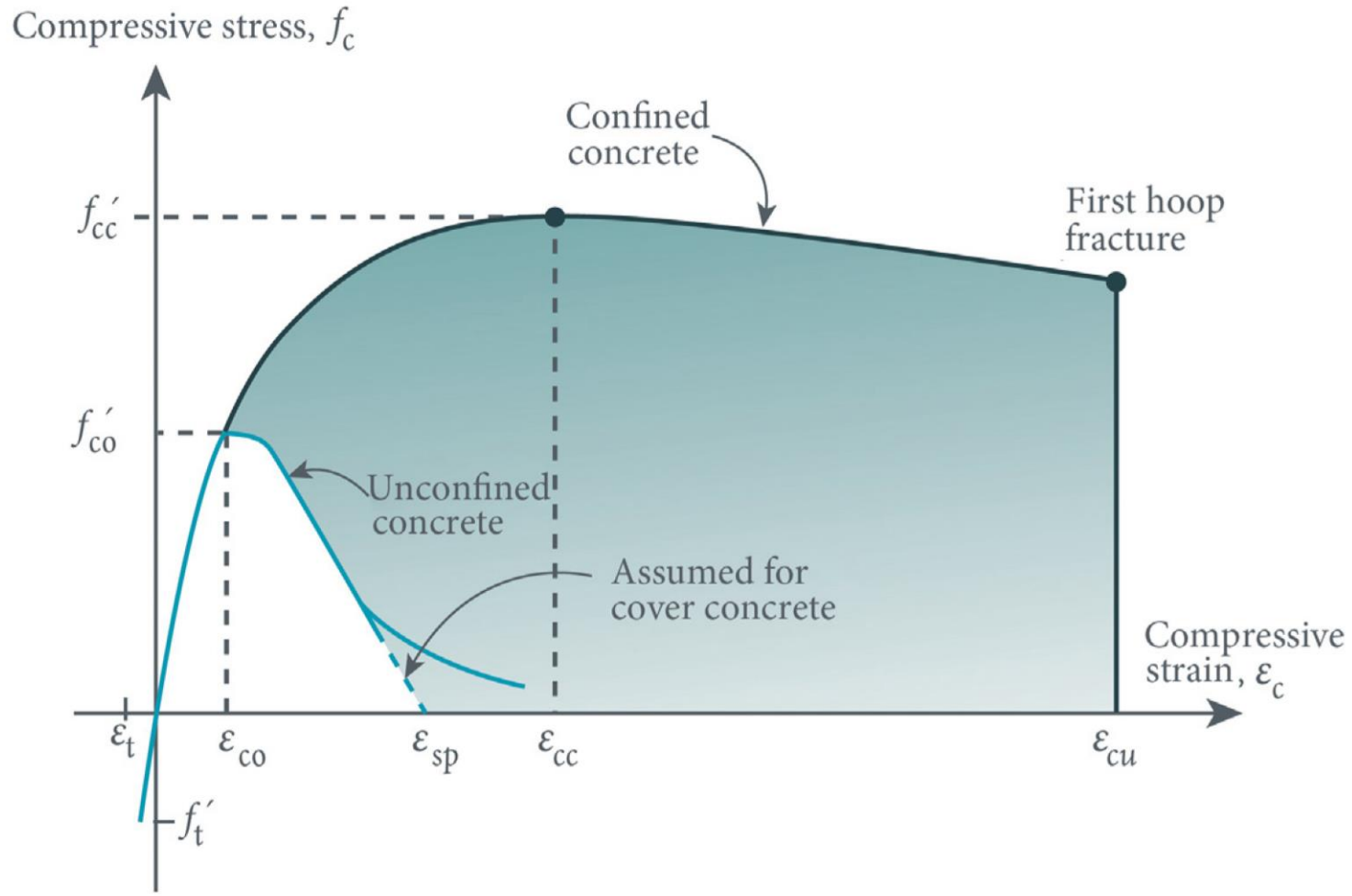
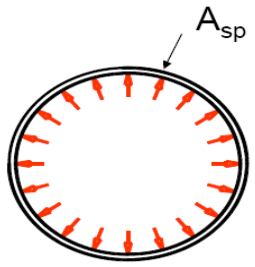
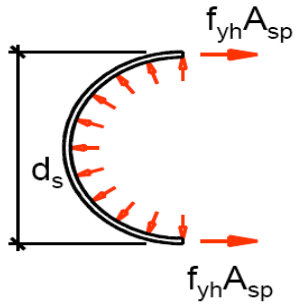


Figure 6.19 The Mander's model for confined concrete (Mander et al., 1988).

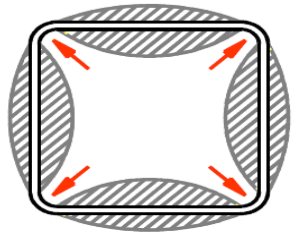
Confinements



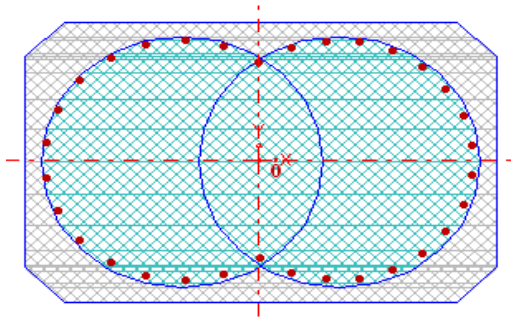
Confinement from spiral or circular hoop

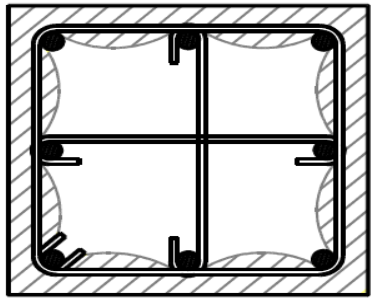


Forces acting on 1/2 spiral or circular hoop

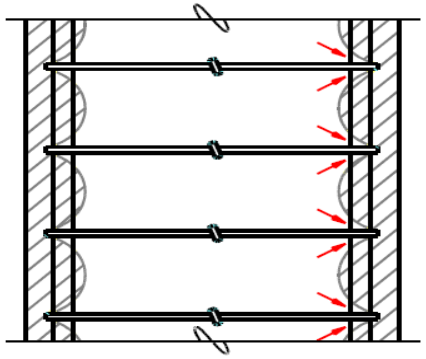


Confinement from square hoop

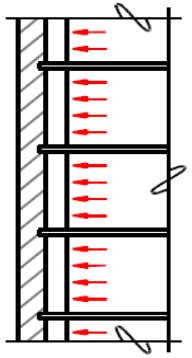




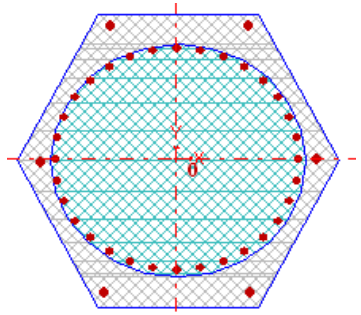
Rectangular hoops with cross ties



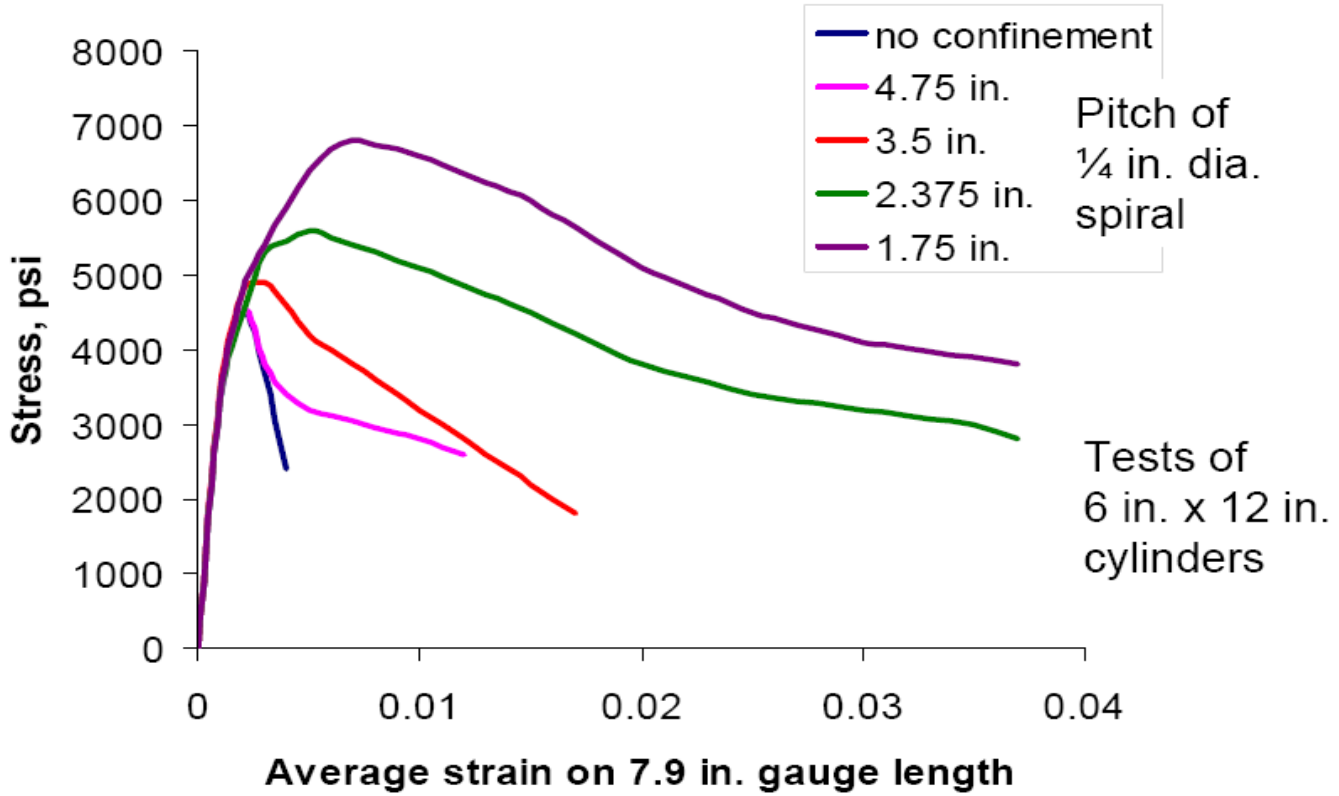
Confinement by transverse bars



Confinement by longitudinal bars

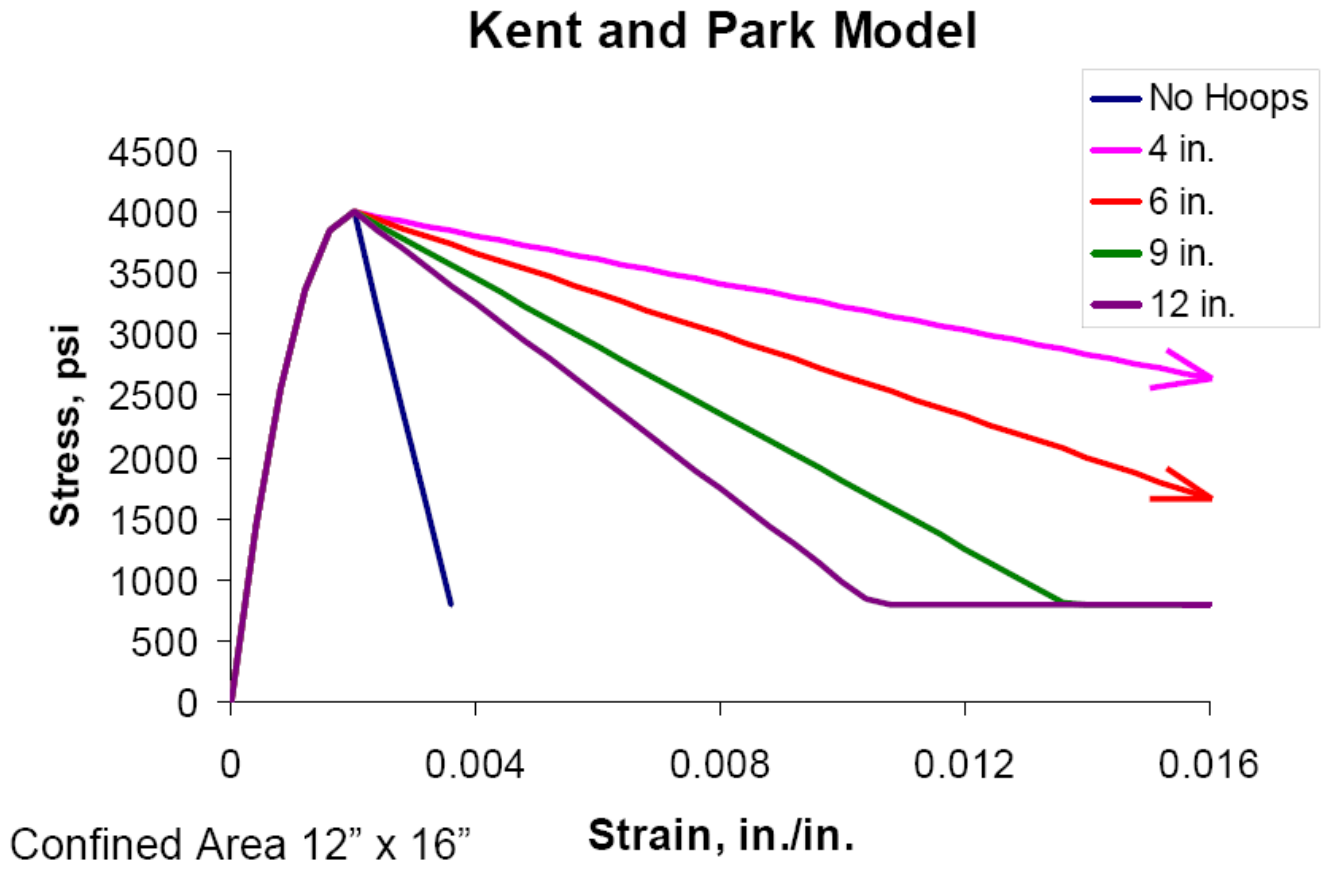


Concrete Behavior and Confinement



Confined Concrete Stress-Strain Behavior

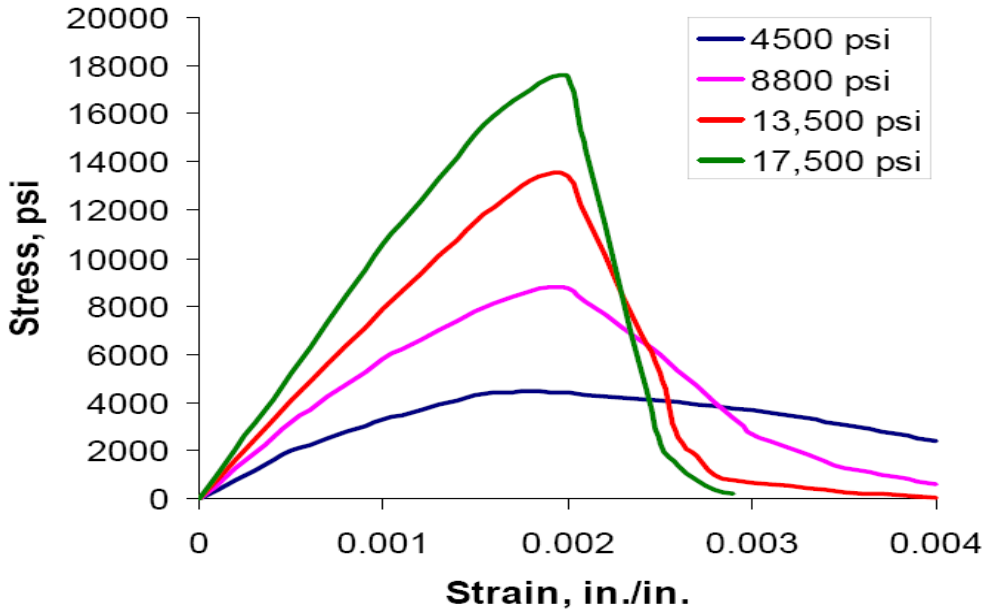
Concrete Behavior and Confinement



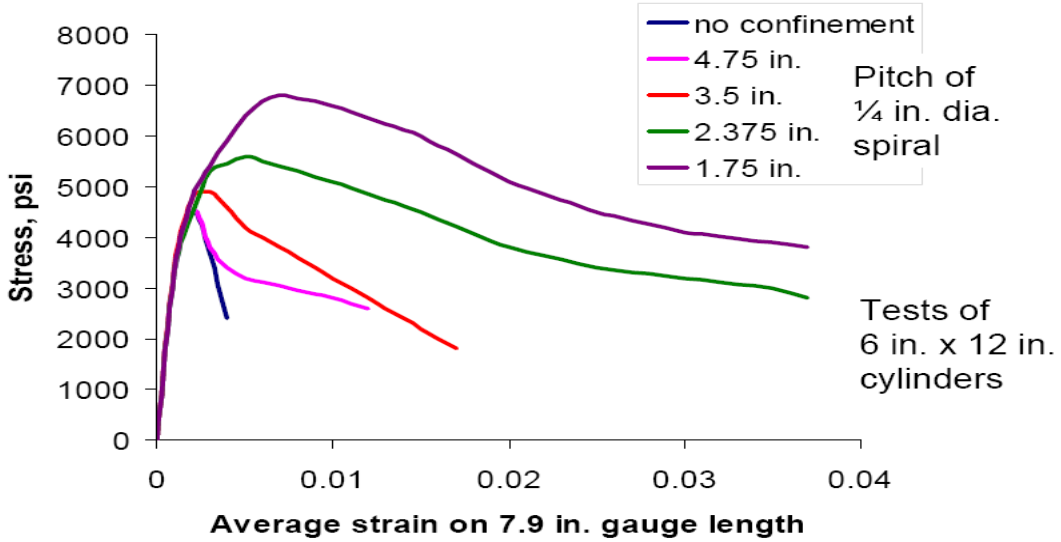
Idealized Stress-Strain Behavior of Confined Concrete

Comparison of Confine and Un-Confined Concrete

Unconfined Concrete Stress-Strain Behavior



Confined Concrete Stress-Strain Behavior



Example 1: Simple Concrete Parametric Stress-Strain Curve

The compression portion of the Simple concrete parametric stress-strain curve consists of a parabolic portion and a linear portion. The following parameters define the Simple concrete parametric stress-strain curve.

ε = Concrete strain

f = Concrete stress

f'_c = Concrete compressive strength

ε'_c = Concrete strain at f'_c

ε_u = Ultimate concrete strain capacity

The concrete Simple parametric stress-strain curve is defined by the following equations:

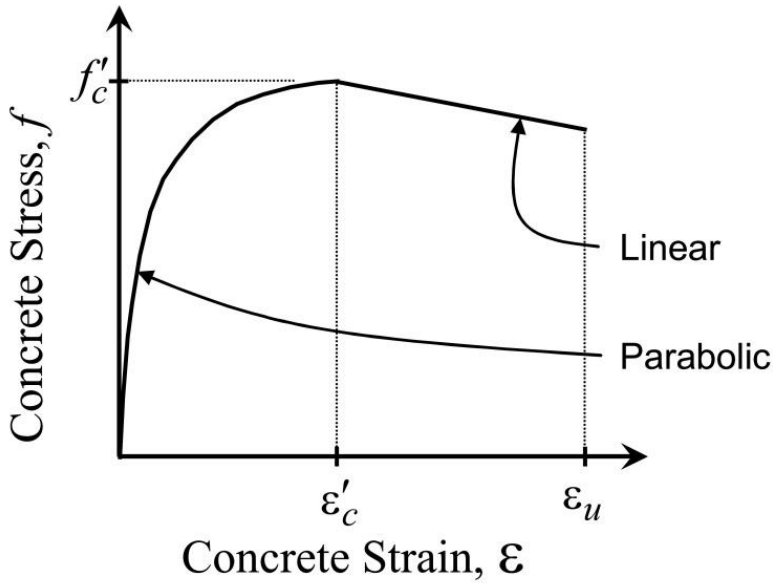
For $\varepsilon \leq \varepsilon'_c$ (parabolic portion),

Example 1: Simple Concrete Parametric Stress-Strain Curve

$$f = f'_c\left\{2\left(\frac{\varepsilon}{\varepsilon'_c}\right) - \left(\frac{\varepsilon}{\varepsilon'_c}\right)^2\right\}$$

For $\varepsilon'_c < \varepsilon \leq \varepsilon_u$ (linear portion),

$$f = f'_c\left\{1 - 0.2\left(\frac{\varepsilon - \varepsilon'_c}{\varepsilon_u - \varepsilon'_c}\right)\right\}$$



The tensile yield stress for the Simple concrete curve is taken at $7.5\sqrt{f'_c}$ in psi.

Example 2: Stress-Strain Model for Unconfined Concrete

[Thorenfeldt, Tomaszewicz, and Jensen (1987)]

$$f_c = f_c' \left[\frac{n\left(\frac{\varepsilon_{cf}}{\varepsilon_c'}\right)}{n-1+\left(\frac{\varepsilon_{cf}}{\varepsilon_c'}\right)^{nk}} \right]$$

$$\varepsilon_c' = \left[\frac{f_c'}{E_c}\right]\left[\frac{n}{n-1}\right]$$

$$E_c = 3320\sqrt{f_c'} + 6900$$

$$n = 0.8 + \frac{f_c'}{17} \qquad k = 0.67 + \frac{f_c'}{62}$$

f_c	= stress in concrete at any level
f_c'	= unconfined compressive strength of concrete
ε'_c	= strain at f'_c
ε_{cf}	= strain in concrete at any level
n	= Curve fitting factor (>1) (MPa)
E_c	= initial tangent stiffness of the concrete (MPa)
k	= post-peak decay factor (value must not be less than unity) (MPa)

Example 3: Mander's Stress-Strain Model for Unconfined Concrete

The compression portion of the Mander unconfined stress-strain curve consists of a curved portion and a linear portion. The following parameters define the Mander unconfined concrete stress-strain curve.

ε = Concrete strain

f = Concrete stress

E = Modulus of elasticity

f'_c = Concrete compressive strength

ε'_c = Concrete strain at f'_c

ε_u = Ultimate concrete strain capacity

The Mander unconfined concrete stress-strain curve is defined by the following equations:

For $\varepsilon \leq 2\varepsilon'_c$ (curved portion),

$$f = \frac{f'_c x r}{r - 1 + x^r}$$

where

$x = \varepsilon/\varepsilon'_c$

$$r = \frac{E}{E - (f'_c/\varepsilon'_c)}$$

Example 3: Mander's Stress-Strain Model for Unconfined Concrete

For $2\varepsilon'_c < \varepsilon \leq \varepsilon_u$ (linear portion),

$$f = \left(\frac{2f'_c r}{r - 1 + 2^r}\right)\left(\frac{\varepsilon_u - \varepsilon}{\varepsilon_u - 2\varepsilon'_c}\right)$$

where r is as defined previously for the curved portion of the curve.

The tensile yield stress for the Mander unconfined curve is taken at $7.5\sqrt{f'_c}$ in psi.

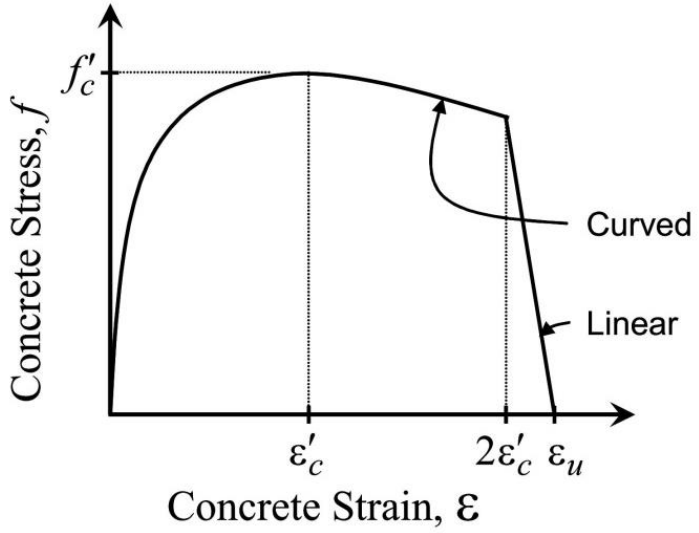


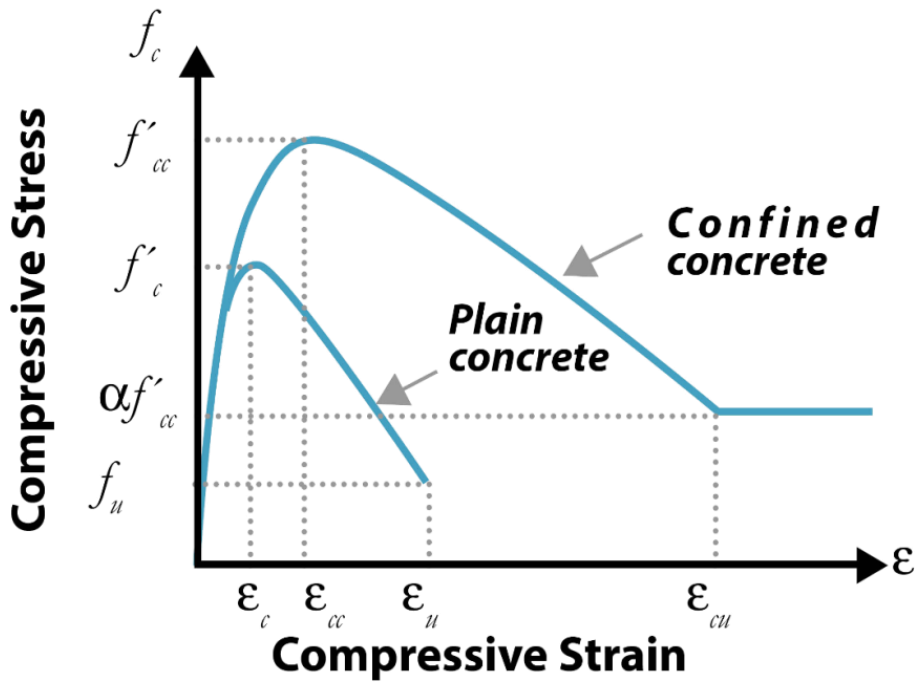
Figure 7 Mander Unconfined Concrete Stress-Strain Curve

Mander's Stress-Strain Model (1988) for Confined Concrete

Reading Assignment 1

Theoretical stress-strain model for confined concrete by J. B. Mander, M. J. N. Priestley, and R. Park

Journal of Structural Engineering, Vol. 114, No. 8, August, 1988



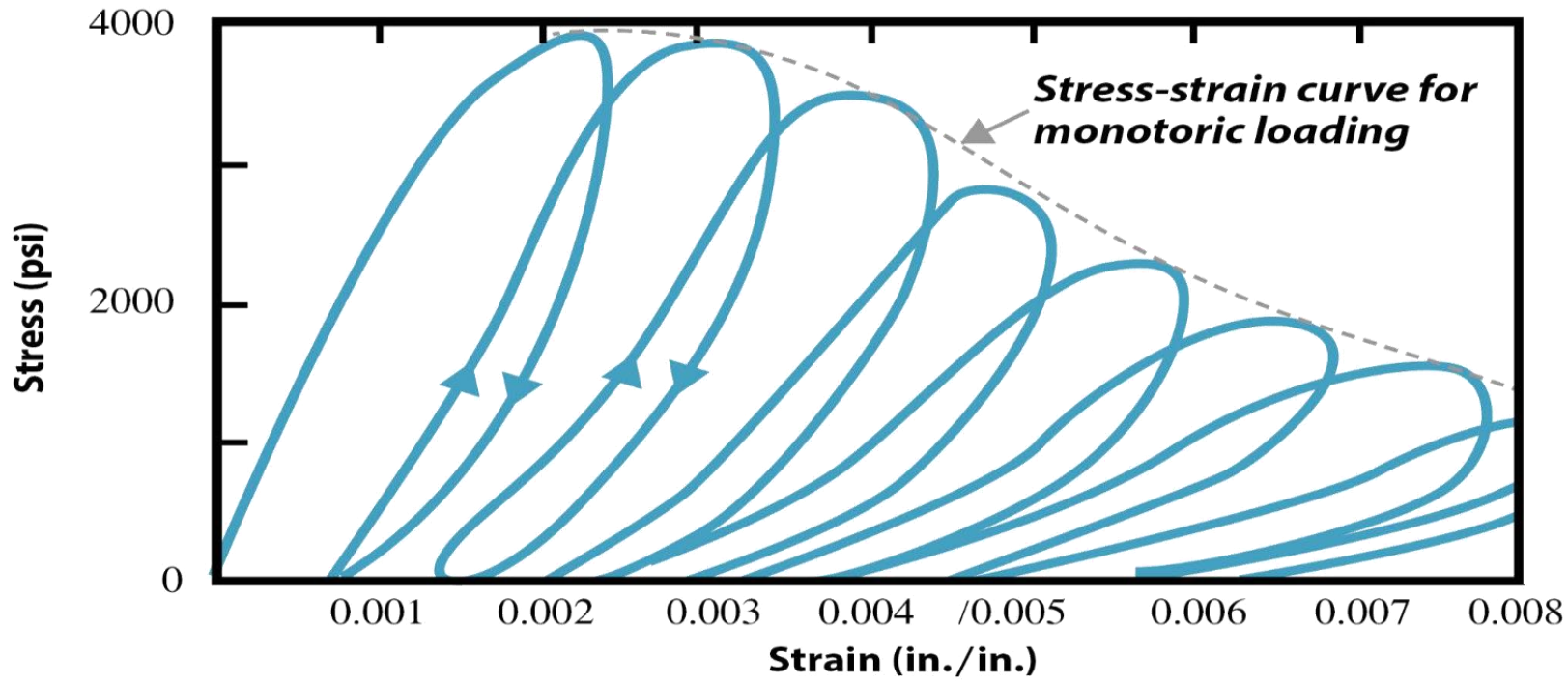
Sheikh and Uzumeri Stress-Strain Model (1982) for Confined Concrete in Tied Columns

Reading Assignment 2

Analytical Model for Concrete Confinement in Tied Columns

Journal of the Structural Division, December 1982.

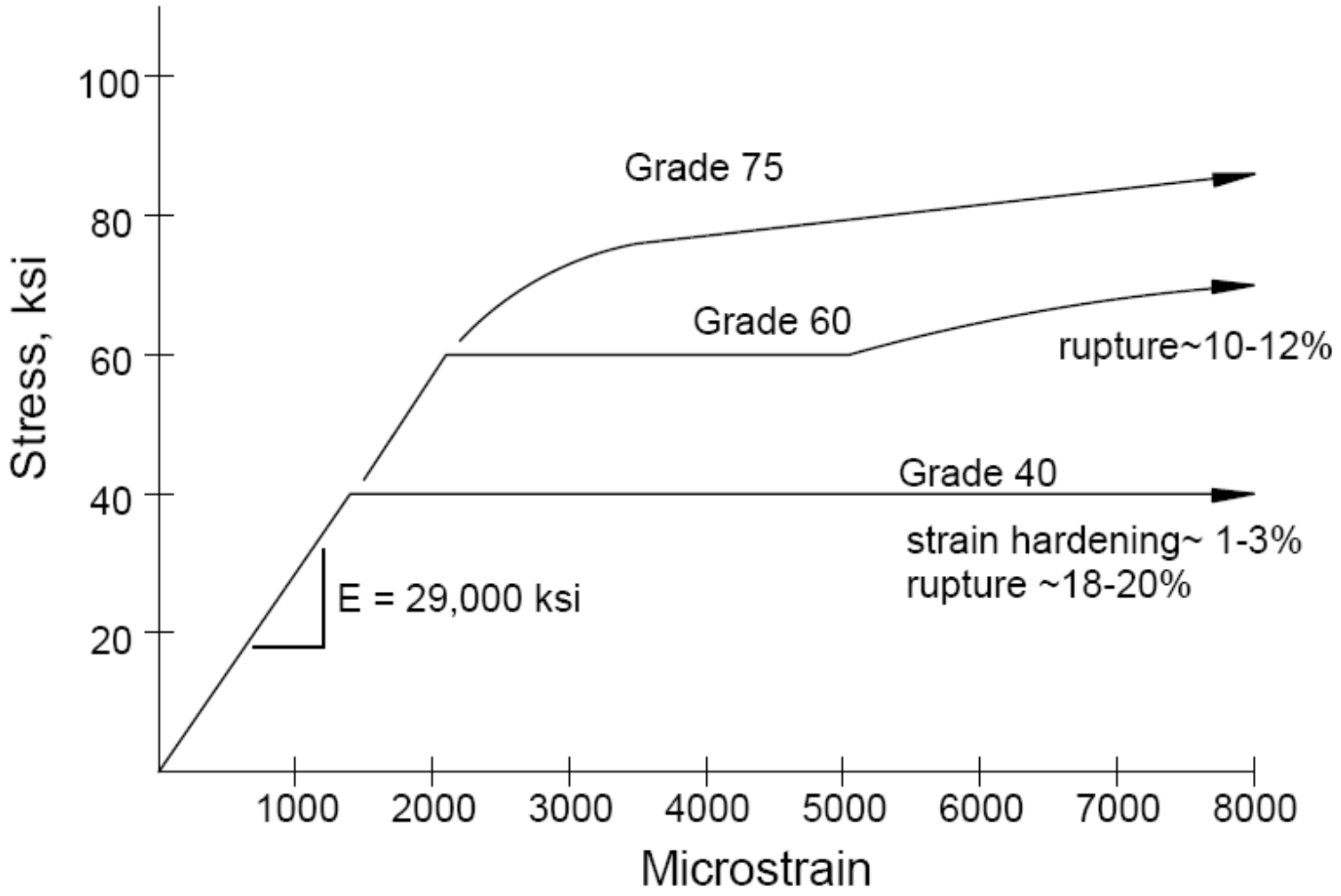
Cyclic Stress Strain Relationship for Concrete



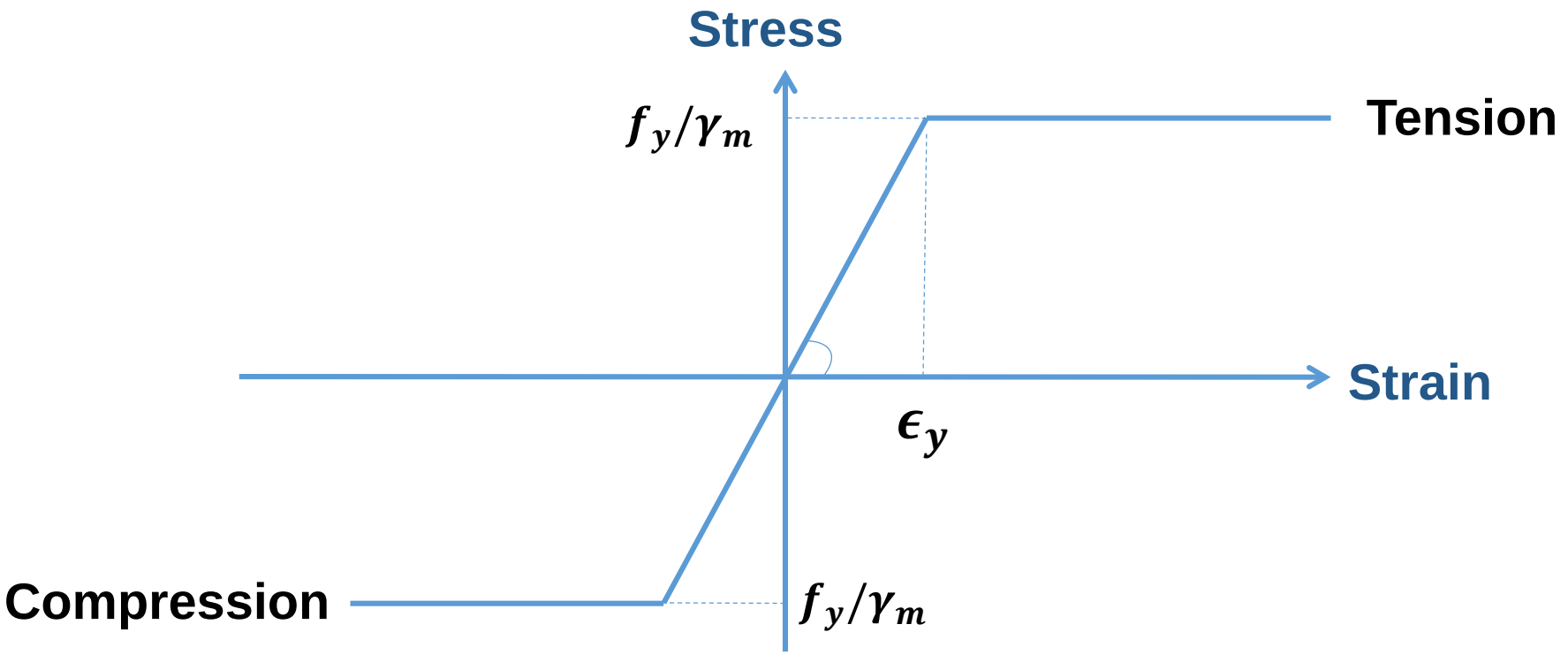
Reference: James G. Macgregor
Reinforced Concrete: Mechanics and Design, 3rd Edition

Stress-strain Curves of Steel

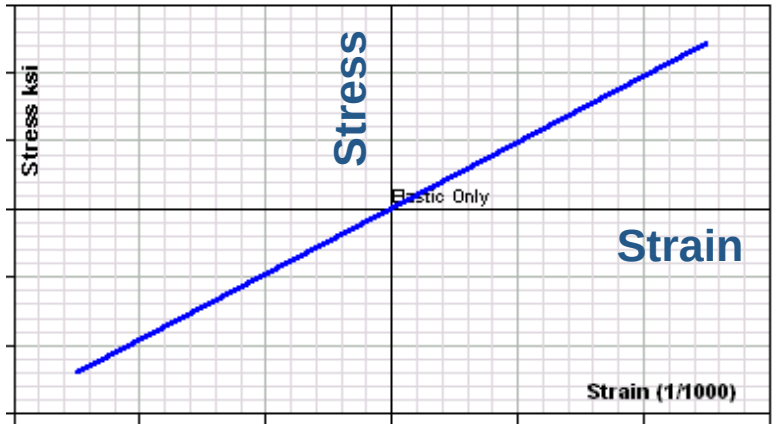
Steel Reinforcement Behavior



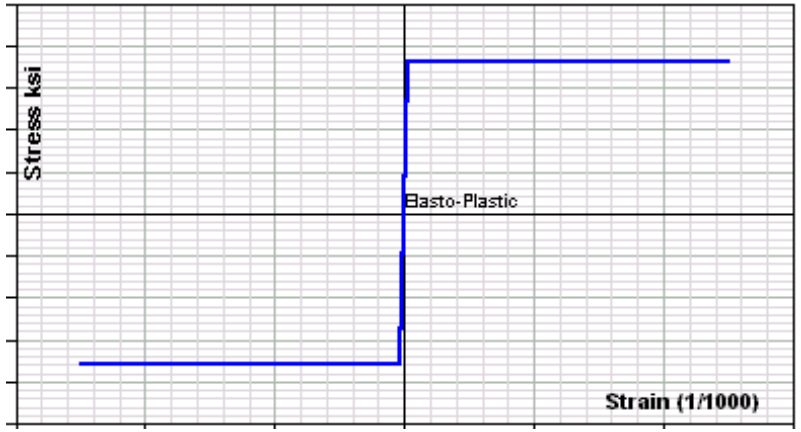
Steel Stress Strain Curve – BS 8110



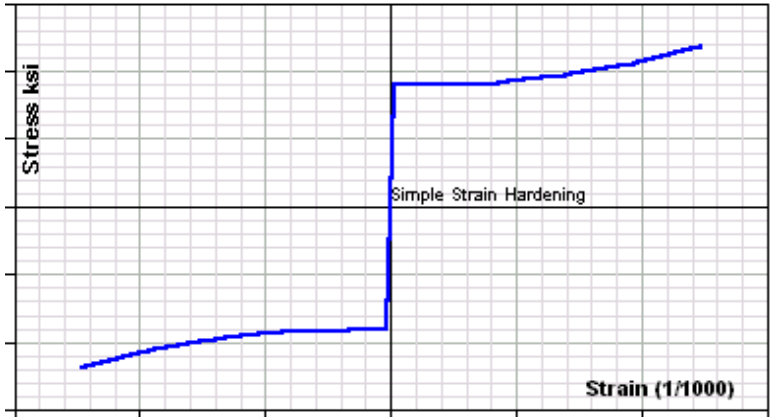
Various Steel Models



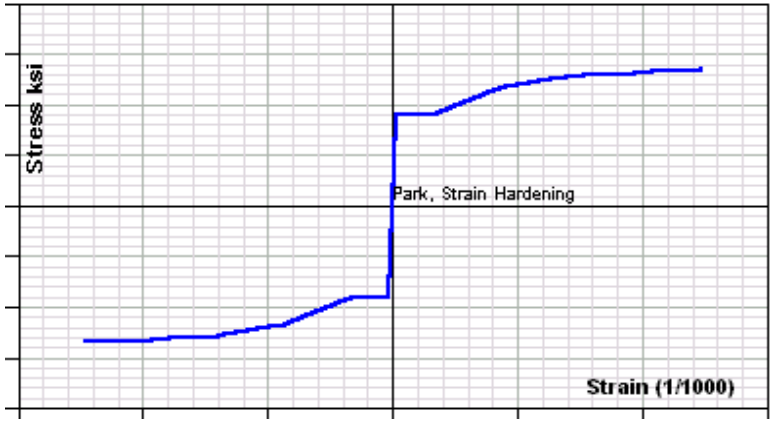
Linear - Elastic



Elasto-Plastic



Strain Hardening - Simple



Strain Hardening Park

Bilinear (Elastoplastic) Steel Stress Strain Relationship

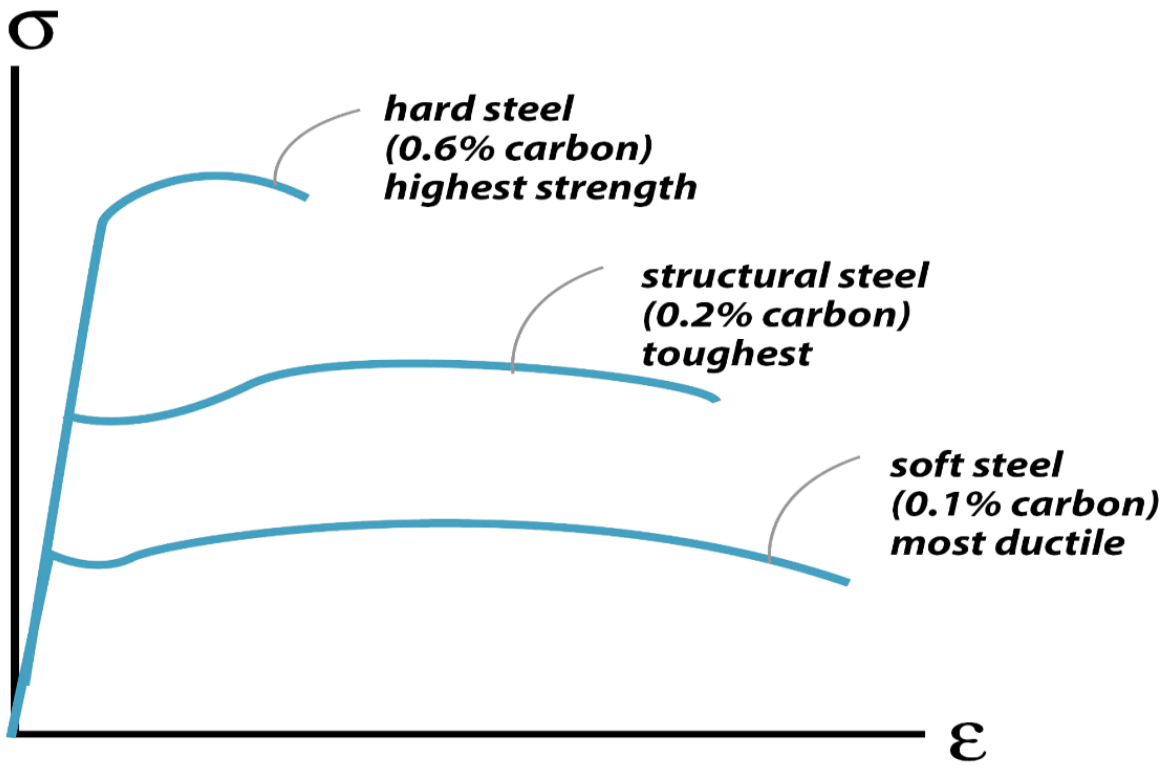
- The reinforcing steel is assumed to be elastic until the yield strain ε_y, and perfectly plastic for strains between ε_y and the hardening strain or until the limit strain ε_{su}, represented by three linear relationship. However, two linear stress-strain relationship is still being used, and simply expressed as

$$f_s = E_s \varepsilon_s \quad \text{, for} \quad \varepsilon_s \leq \varepsilon_y$$

$$f_s = f_y \quad \text{, if} \quad \varepsilon_s > \varepsilon_y$$

  - f_s= stress in reinforcing steel at any level y due to εs
  - E_s = 200,000 MPa, modulus of elasticity of reinforcing steel

Steel Stress Strain Relationship

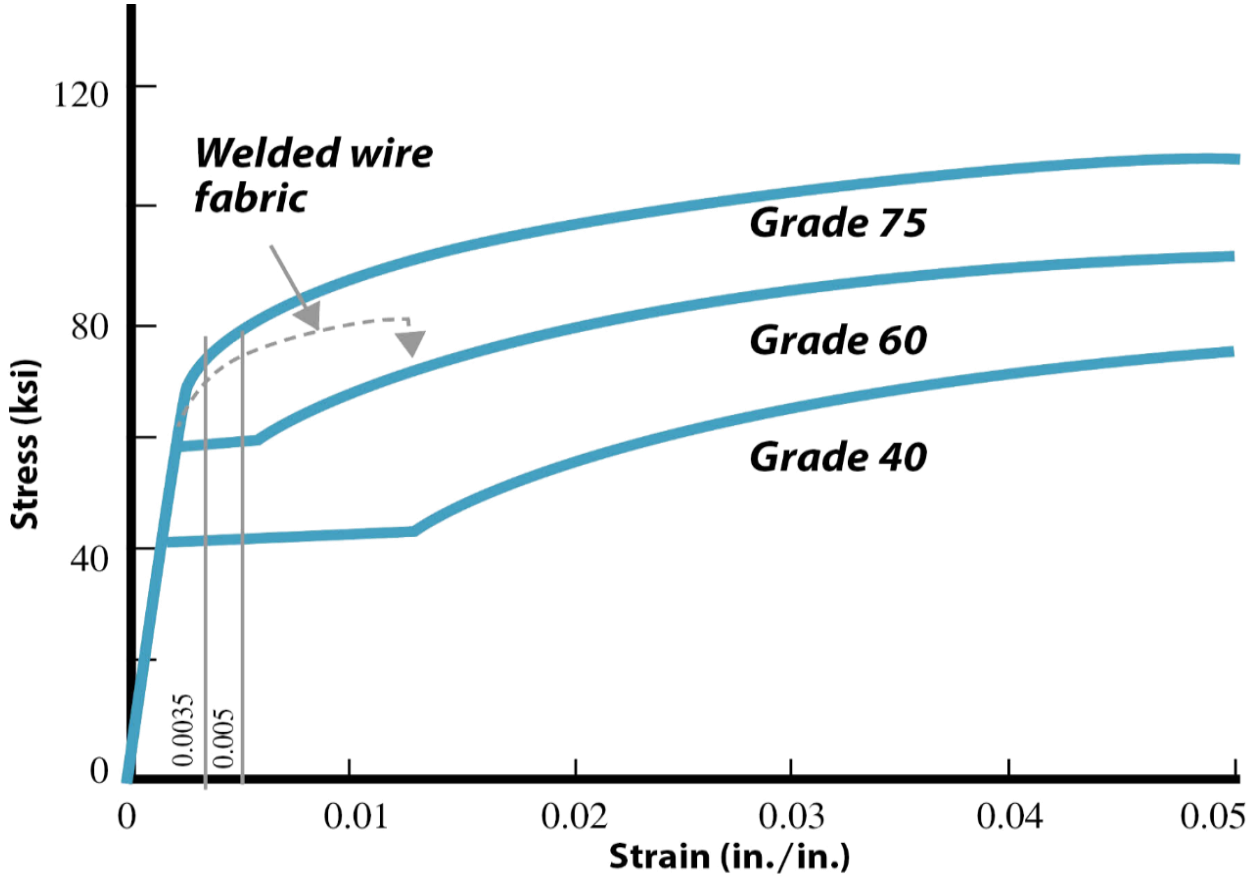


Steel: Stress-strain diagrams for different steels (Hibbeler, 1997)

Reference: James G. Macgregor

Reinforced Concrete: Mechanics and Design, 3rd Edition

Steel Stress Strain Relationship



Reference: James G. Macgregor
Reinforced Concrete: Mechanics and Design, 3rd Edition

Standard Reinforcing Bars (US Designation)

			Nominal Dimensions*	
Bar Size Designation No.	**Grades**	**Weight (lb/ft)**	**Diameter (in.)**	**Cross-Sectional Area (in^2.)**
3	40, 60	0.376	0.375	0.11
4	40, 60	0.668	0.500	0.20
5	40, 60	1.043	0.625	0.31
6	40, 60	1.502	0.750	0.44
7	60	2.044	0.875	0.60
8	60	2.670	1.000	0.79
9	60	3.40	1.128	1.00
10	60	4.30	1.270	1.27
11	60	5.31	1.410	1.56
14	60	7.65	1.693	2.25
18	60	13.60	2.257	4.00

**** The nominal dimensions of a deformed bars are equivalent to those of a plan round bar having the same weight per foot as the deformed bar.***

Steel strain due to thermal changes

- At low temperatures, steel becomes harder and more brittle while it becomes softer and more ductile when the temperature rises. Although the thermal expansion for steel is actually 6.5x10-6/0F, it is conventional to use a value of 6x10-6/0F for both concrete and reinforcement (Collins and Mitchell, 1991)

$$\varepsilon_s = \pm \alpha_{sb} \Delta T$$

$\alpha_{sb} = 6x10^{-6} / ^0 F$, coefficient of thermal expansion

ΔT = changes in temperature in ^0F

Temperature Dependent Properties

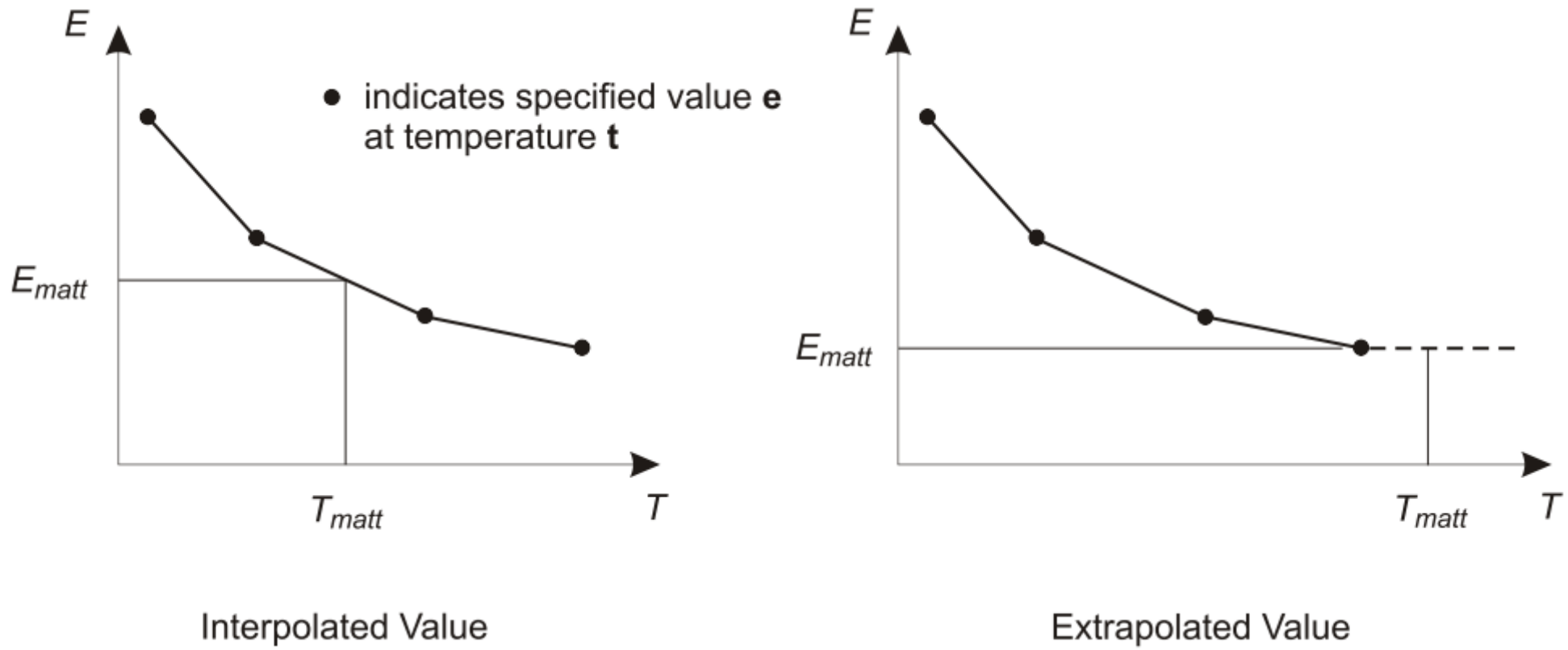
- These properties are given at **a series of specified material temperatures, *t***.

- Properties at other temperatures are obtained by **linear interpolation** between the two nearest specified temperatures.

- Properties at temperatures outside the specified range use the properties at the nearest specified temperature.

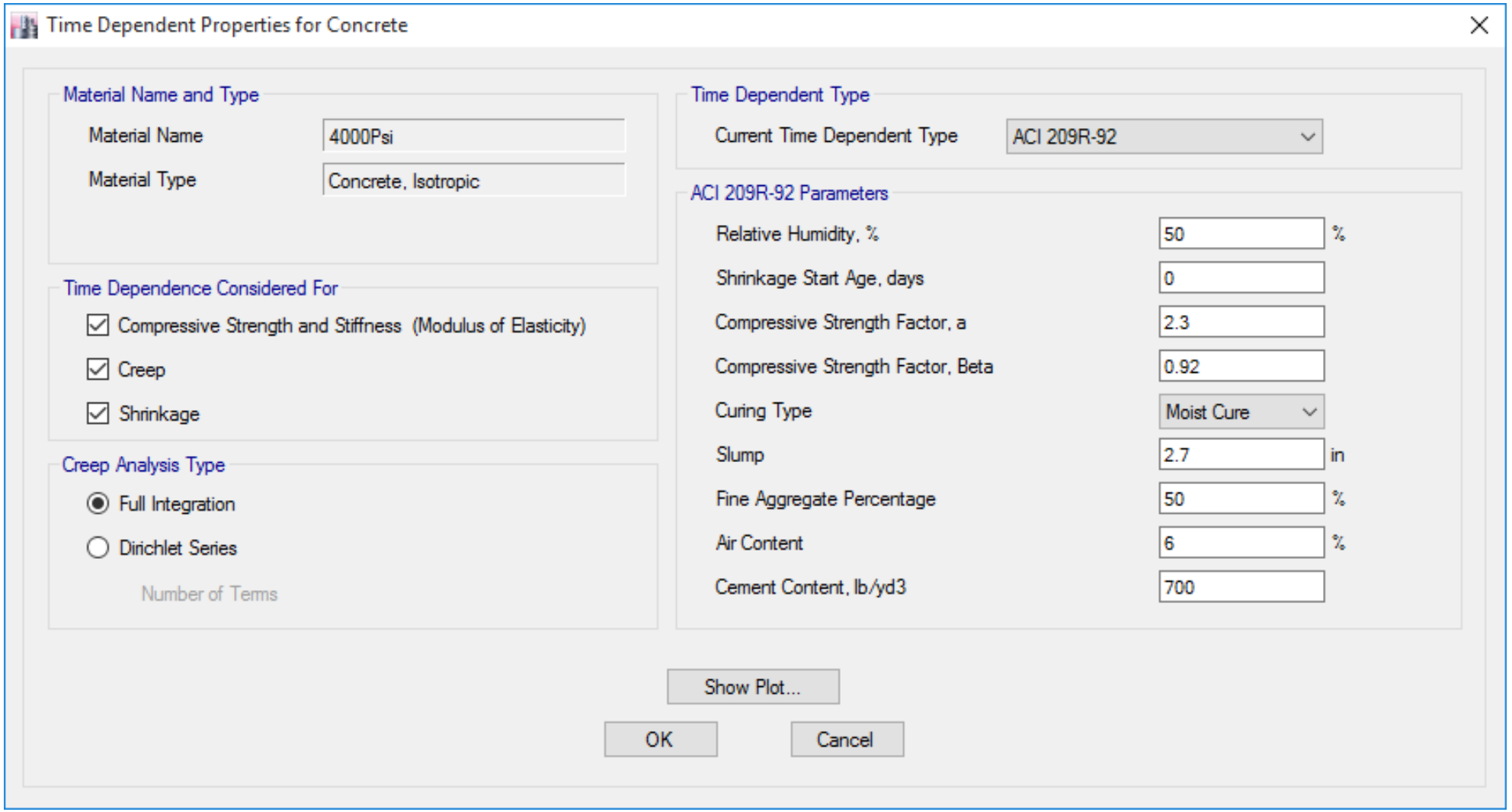
Special Properties of Concrete

Temperature Dependent Properties

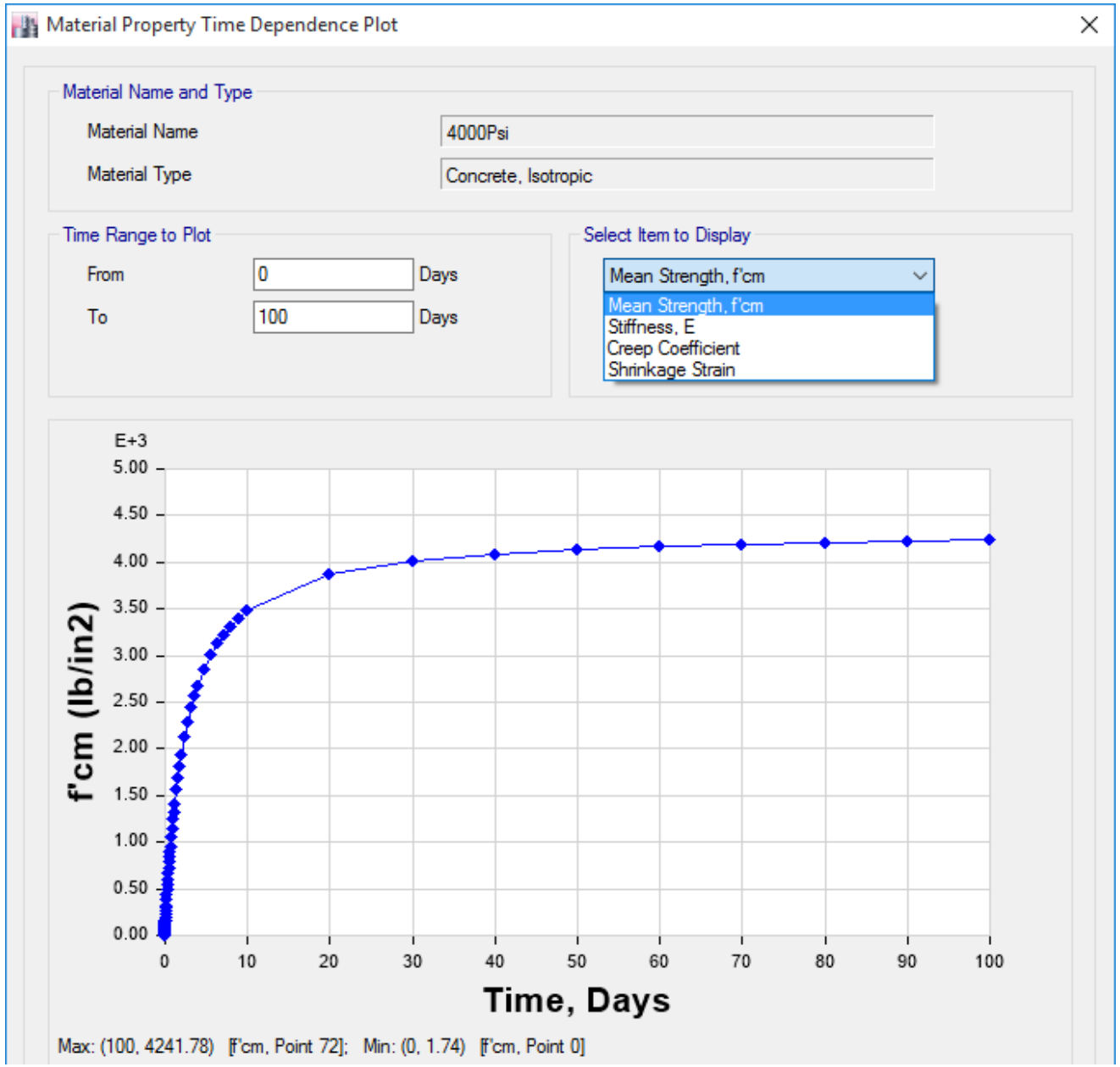


Determination of Property E_{matt} at Temperature T_{matt} from Function E(T)

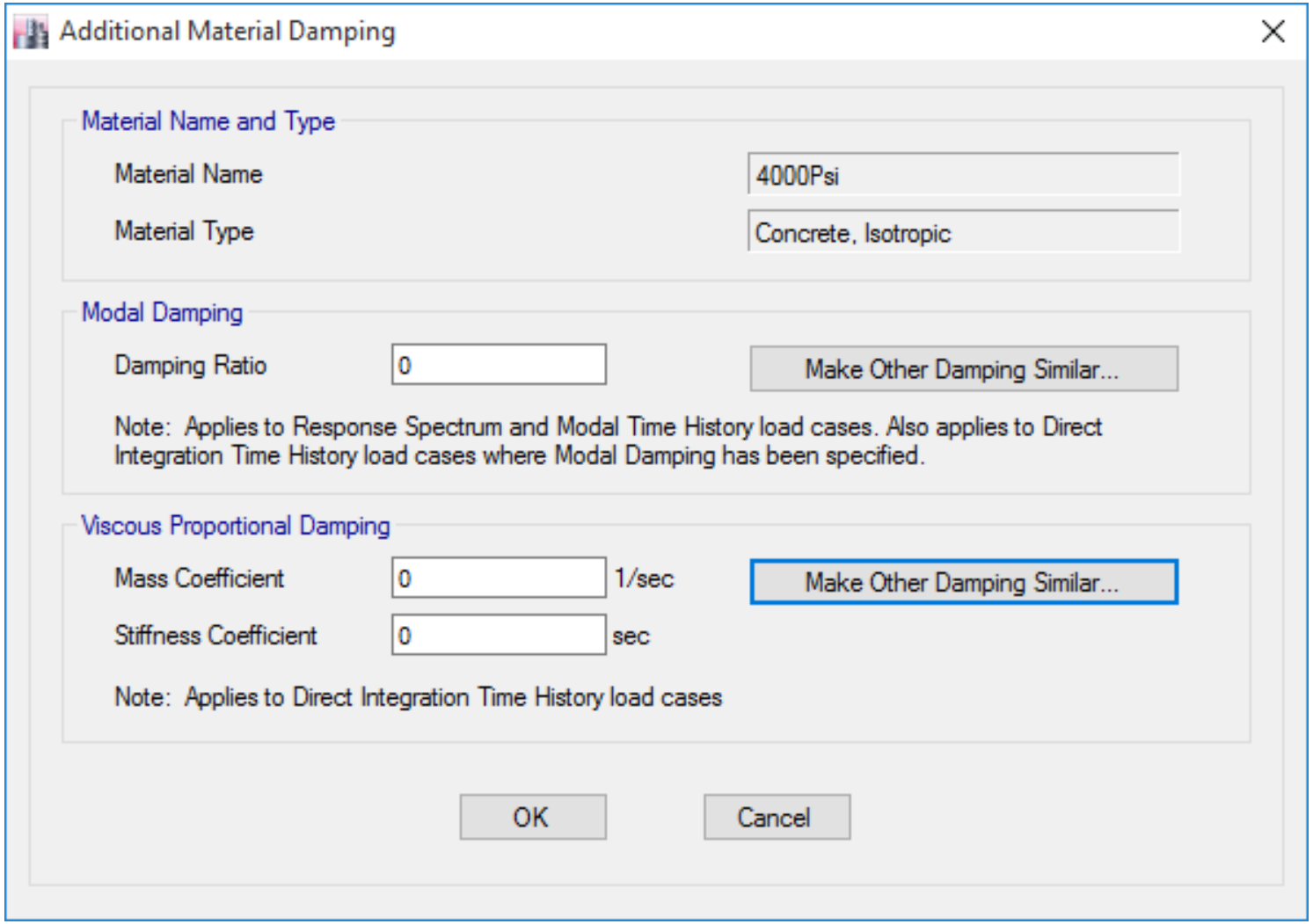
Temperature Dependent Properties



Temperature Dependent Properties



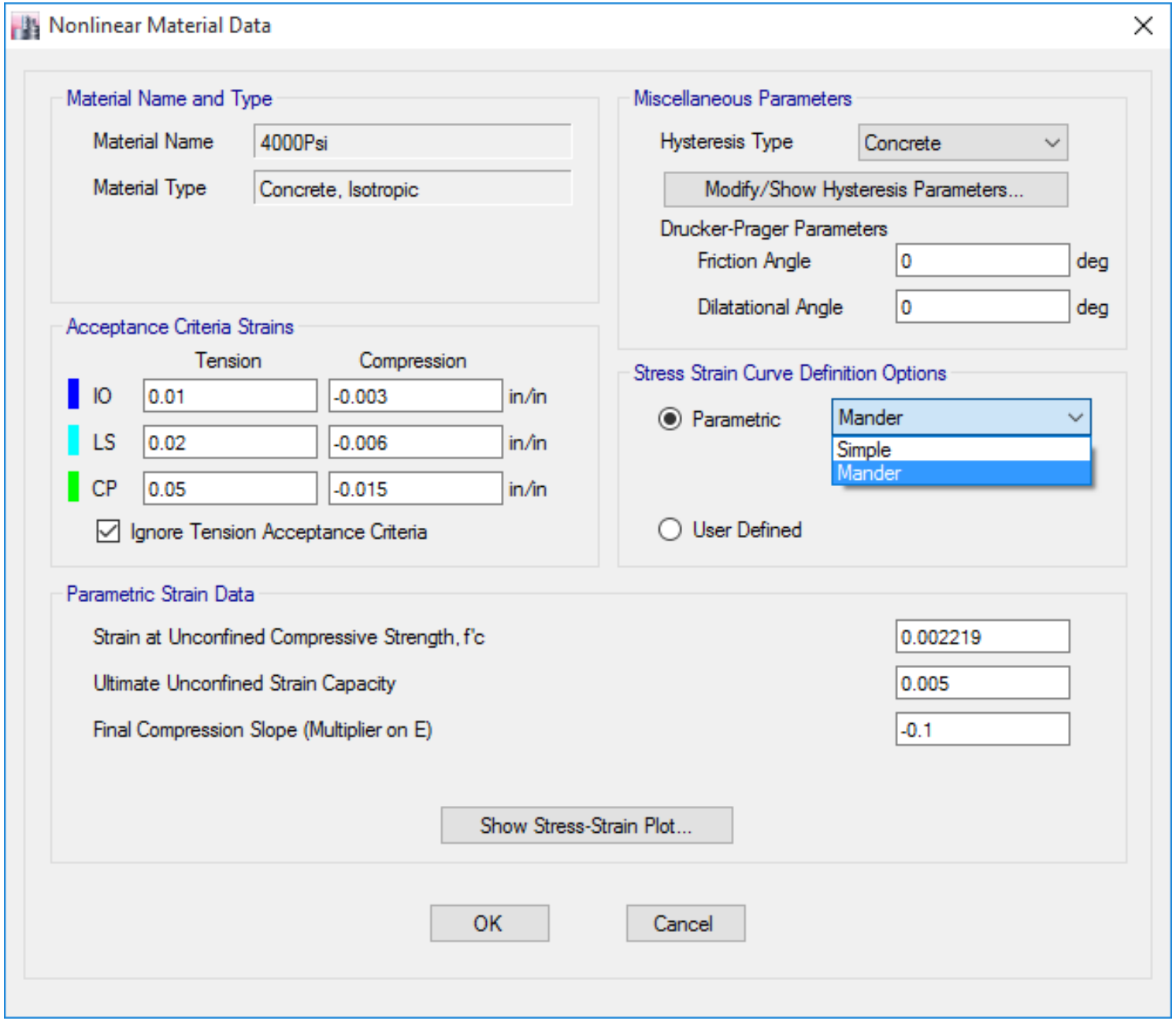
Material Damping



Nonlinear Material Properties

Used in the nonlinear modeling of elements using the

- **Fiber Hinges**
- **Layered Shell Element**



The Role of Material Properties in Calculating Internal Stresses

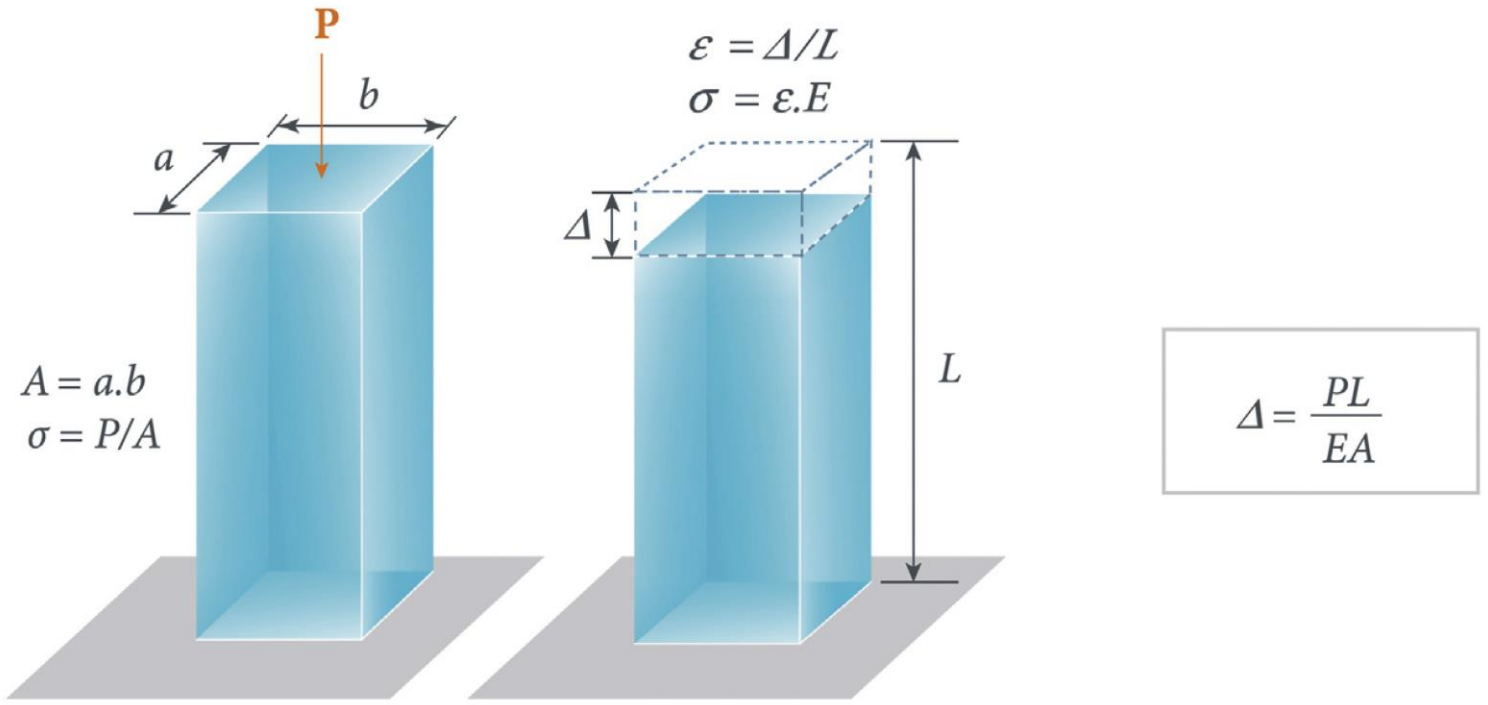


Figure 3.1 (A)Stress can be computed as direct result of action, irrespective of the material or, (B) due to deformation, using material properties. (C) Both methods can be related by the action–deformation relationship for certain cases.

The Role of Material Properties in Cross-sectional Analysis

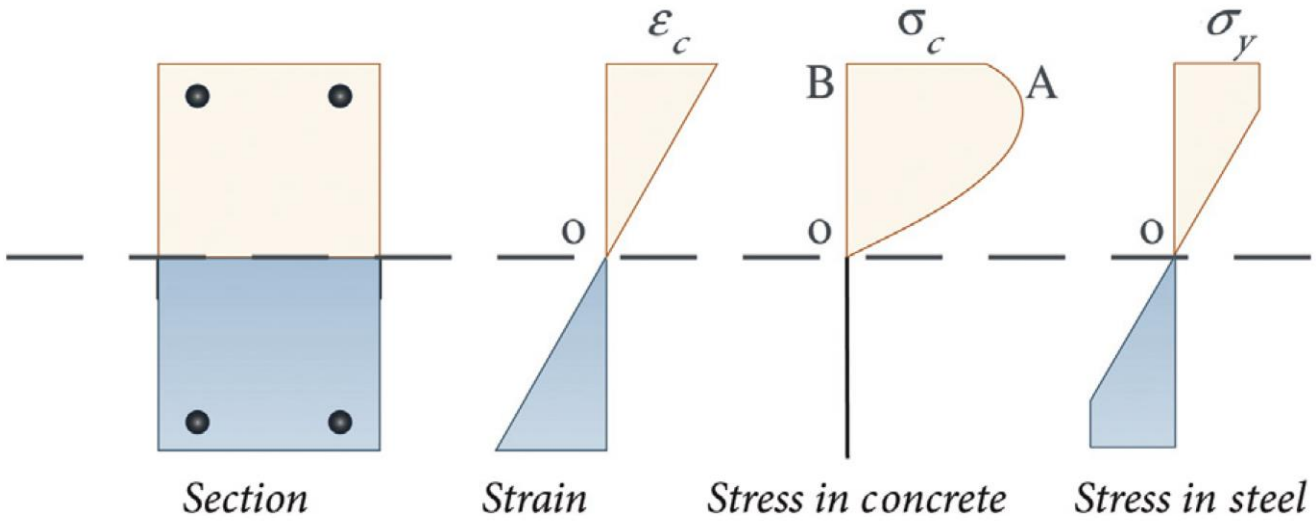


Figure 2.19 The stress–strain curve of material translates to stress diagram on the cross-section.

Conversion of Original Stress Block into an Equivalent Rectangular Stress Block

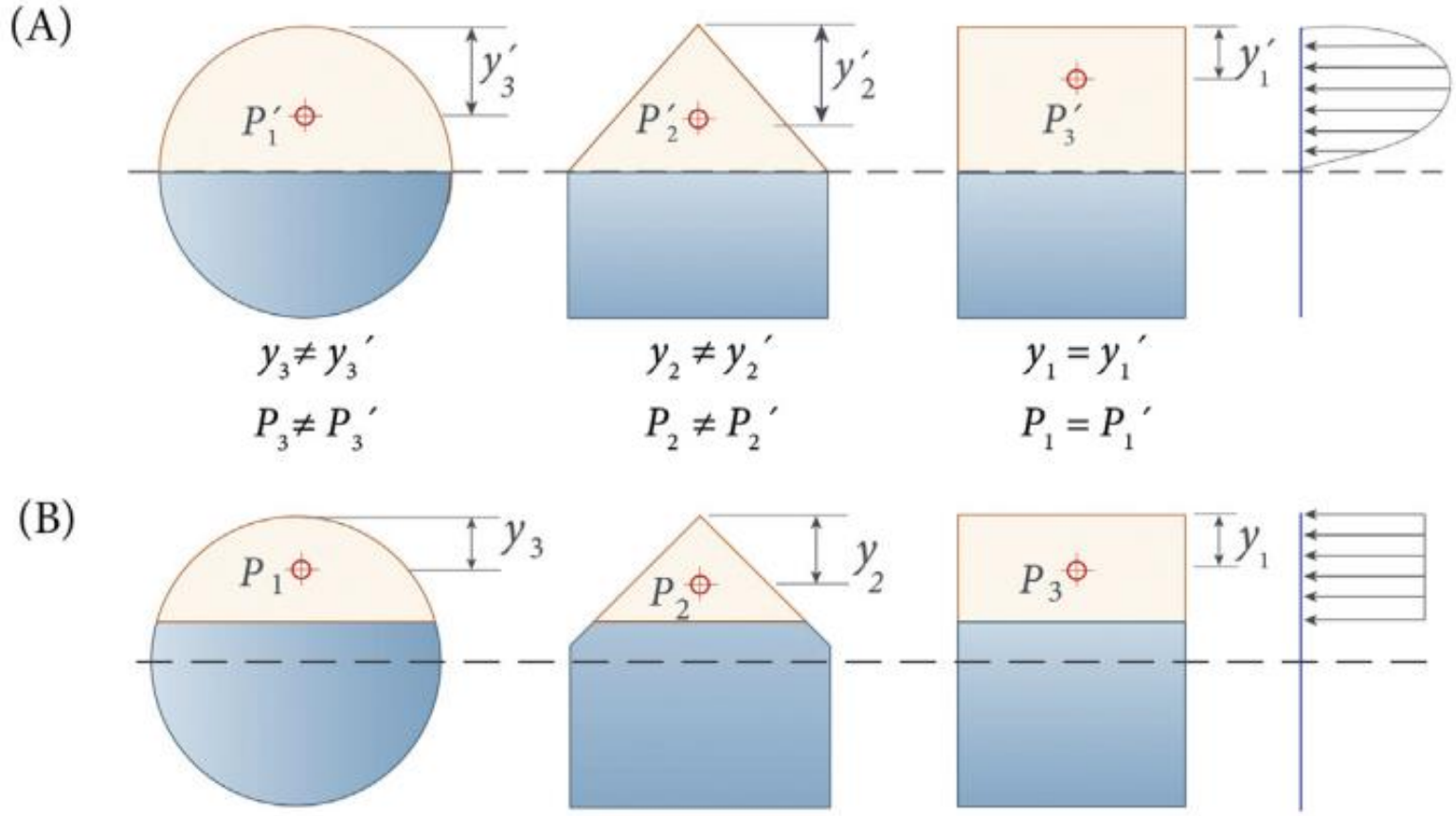


Figure 2.21 The simplified stress may not give the same results as original stress block: (A) original stress block; (B) simplified stress block.

The Consideration of Complete Stress-strain Curve of Concrete

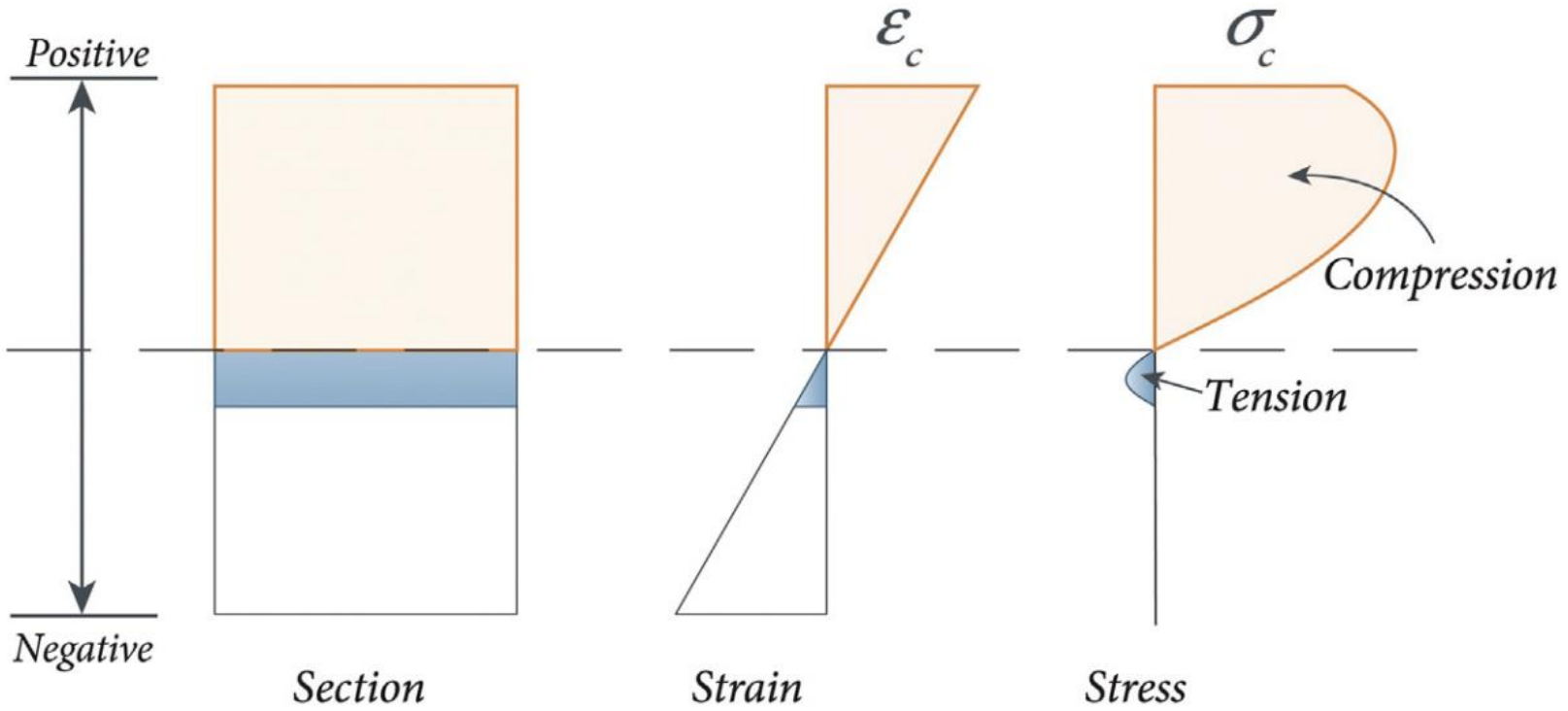


Figure 2.22 Tension capacity of concrete can be considered by including the tension part of the stress–strain curve.

Assignment 1

1) Summarize 5 Models for unconfined concrete, with equations and compare their stress-strain curve. Develop a computer program for the parametric analysis and graphical comparison of these models.

2) Summarize 5 Models for confined concrete, with equations and compare their stress-strain curve for a column with following details.

   [500 x 500 mm, f_c'= 40 MPa, 16 vertical bars of dia. 25 mm (f_y= 420 MPa), and confined by hoops of dia. 12 @ 200 mm (f_y= 420 MPA)]

   Develop a computer program for the parametric analysis and graphical comparison of these models.

- Time 2 weeks

Thank You